

The phenomeno-logical robot



Jean-François Cloutier
Research Fellow
Active Inference Institute

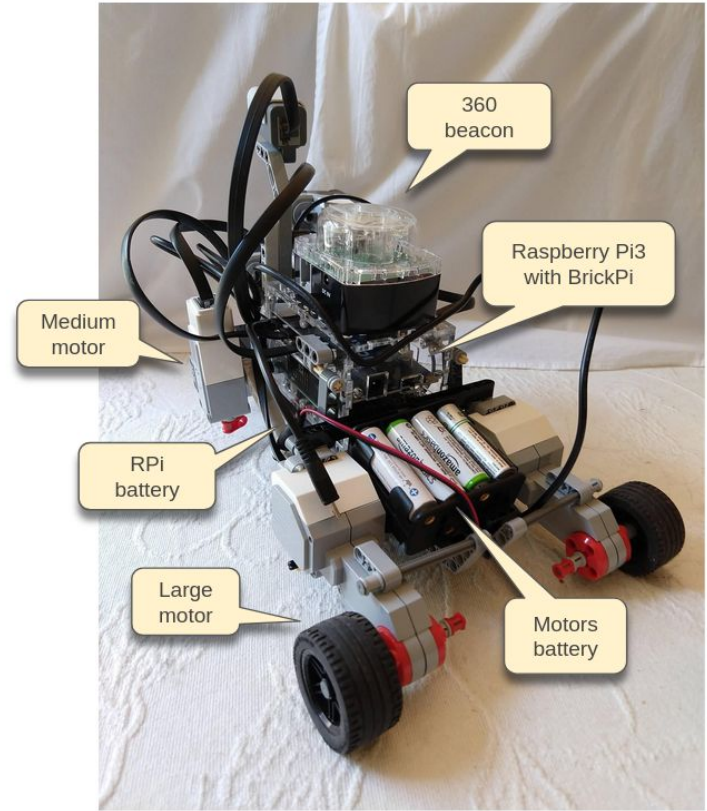
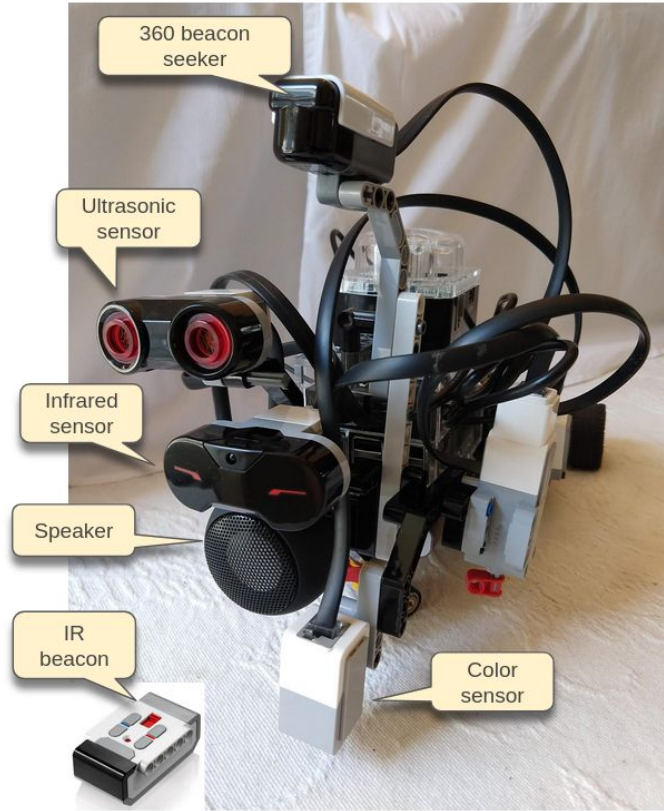
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What does it take, at a minimum, for an autonomous robot to learn to survive in a world it knows initially nothing about?

I am programming a mortal robot to have felt experiences,
make sense of them, and grow the agency it needs to survive

The robot is a rover with Lego EV3 sensors and motors



What this project is not about

This is **not** an *input-output simulation* of *human-level* cognition (what LLMs do)

This is **not** an attempt at a *general model* of cognition

This is **not** a robot *solving* Free Energy Principle equations

But one that *complies* with FEP principles

This is **not** a robot as an extension of its programmer

The robot's agency is grounded in its own experiences

What this project is about

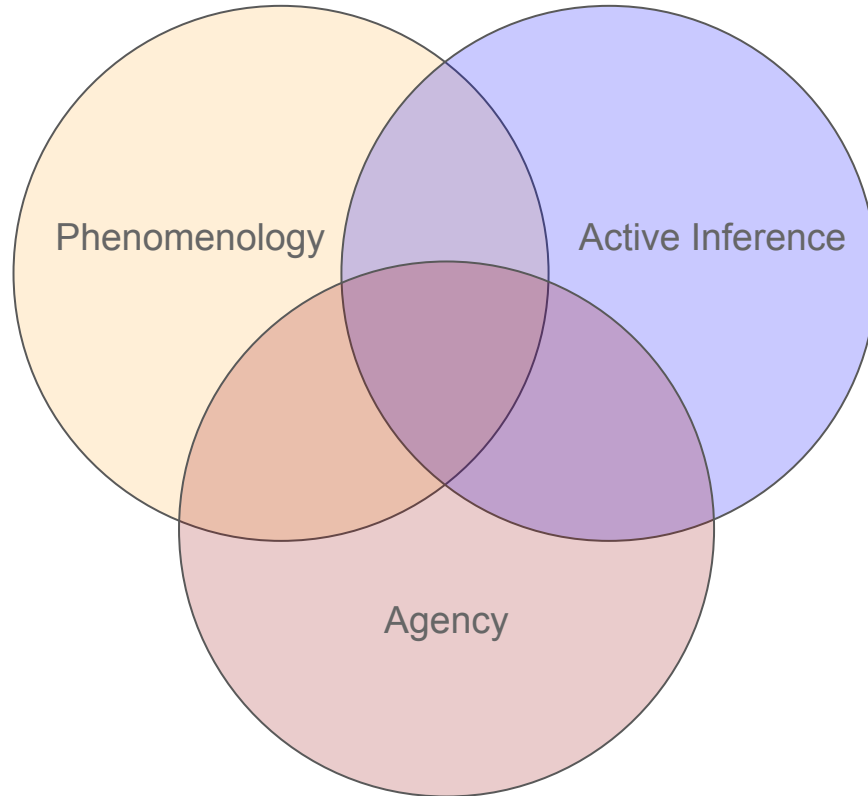
I am *building* an artificial agent

I want to *achieve basal* cognition

Experiences are to be *intrinsically* meaningful to the robot

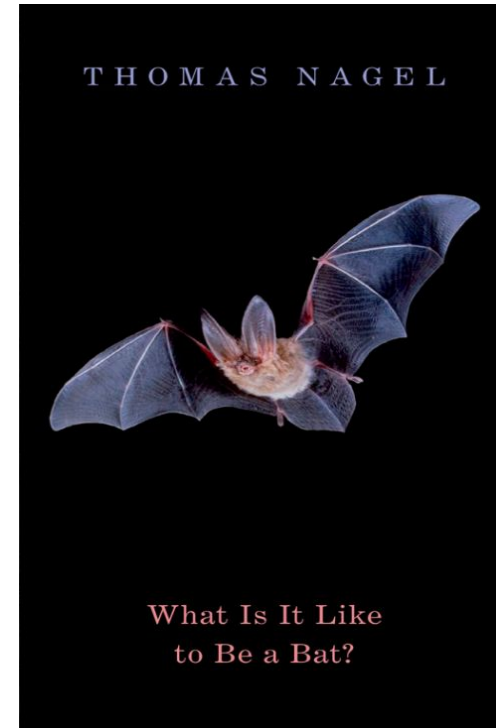
FEP is applied as an *analytical* framework

I am building to understand



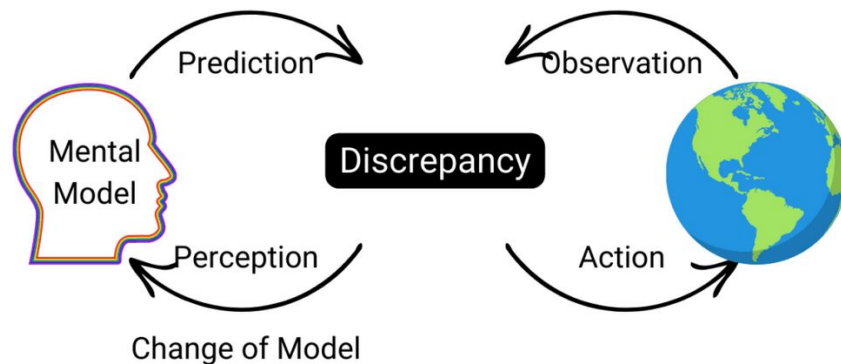
Phenomenology

Phenomenology seeks to determine the *essential properties and structures of experience itself*, and to provide an *objective* account of subjective experience



Active Inference

An enduring being actively infers the causes of its sensory input and shapes its environmental niche to minimize prediction errors



Agency

Agency is the ability of a self to act purposefully to achieve its own goals in an uncertain world it experiences



My project is to program a robot so it learns to survive by experiencing, engaging, and making sense of its world

All on its own

Principles

A self is mortal

Agency implies a self distinct from an environment
it depends on *to survive*

A self is not a “thing” but processes organized to endure

A self is a dynamic organization of processes that constrain and sustain one another

The organization is not fixed but adapts to avoid dissipation

To adapt, a self must continually model itself in its world

A self's overarching goal is to survive

The processes producing the self must adapt their organization to preserve *continuity of being*

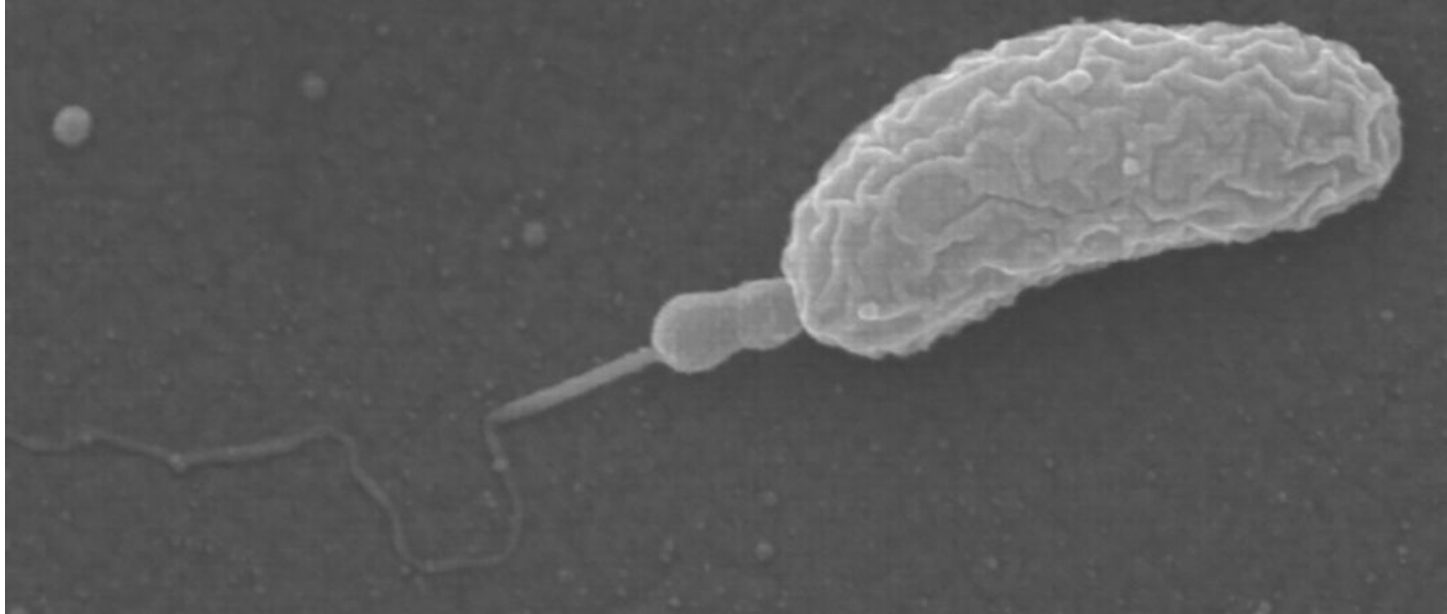
A self strives to remain a self, not necessarily stay the same

Mortality introduces value

What helps the self persist is good for it

What threatens to terminate it is bad for it

Value is subjective



When a bacterium feeds on another, what's good for one is bad for the other

This project defines **agent** as
“A mortal self who chooses how to survive”

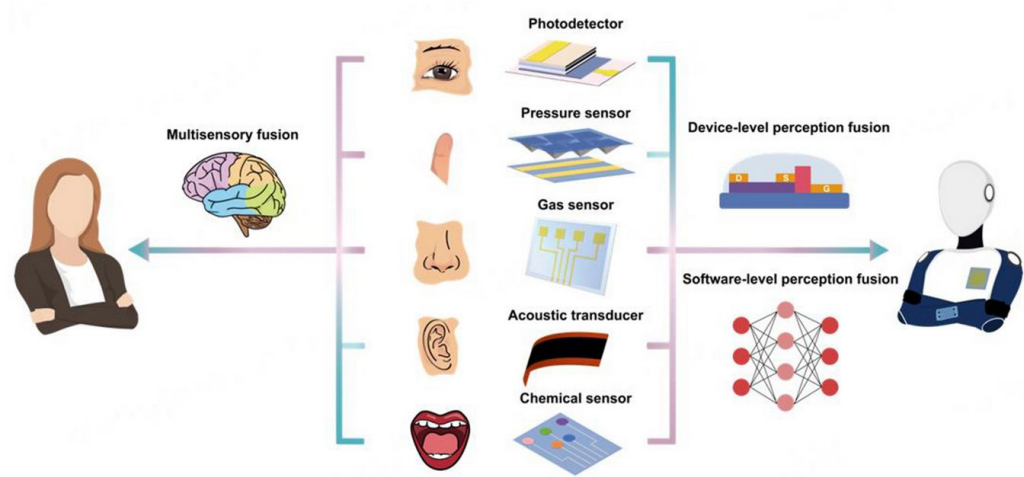


To survive, an agent must make (enough) sense of itself in its environment to predict (well enough) the consequences of its actions or inaction

An agent has felt experiences

The agent's environment impinges on its senses
So do the agent's own actions

An agent constantly processes streams of multimodal sensations



<https://www.oaepublish.com/articles/ss.2023.15>

The agent integrates sensory streams in spacetime to create its own **experiences**

When I bite into an apple, I experience its taste, crunchiness, color, weight etc. at once, in unified moments of awareness

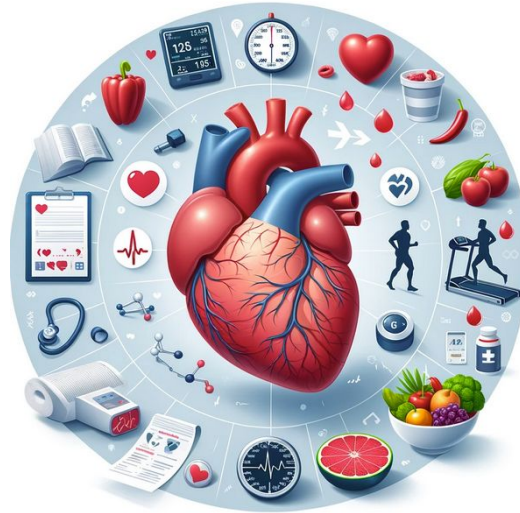
An agent need **not** be aware it is having experiences in order to have them

The agent keeps track of risks to its survival by constantly measuring its **wellbeing**

The wellbeing of an agent is a function of...



Fullness
(how energetic it is)



Integrity (how healthy it is)



Engagement
(how participatory it is)

The agent *correlates* its experiences with fluctuations in its wellbeing

This is how it has *felt* experiences

An agent can have good and bad experiences



An agent is sense making

An agent must make sense of its world if it is to survive in it

An agent makes sense of its experiences by **modeling**
what causes its sensations

It builds a **causal theory** to explain its sensations

The agent uses its causal theory to make **predictions**

The causal theory is a *generative model*

It generates (predicts) incoming sensations including the sensory consequences of the agent's actions

Actual sensations that contradict predicted sensations become **prediction errors**

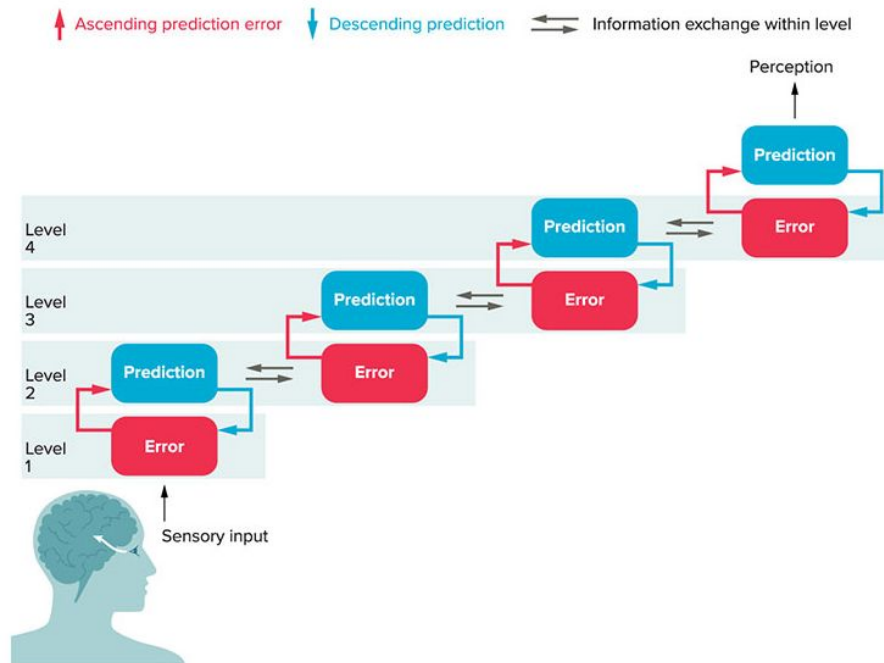
Errors happen because the causal theory is inaccurate or incomplete, or because the world has changed since the causal theory was formulated

Sensations = predictions + prediction errors

What an agent senses are the uncontested predictions it makes plus the prediction errors it receives

What the agent senses is limited to what it can predict

What it can predict changes over time



SOURCE: ADAPTED FROM C.J. HARMER & M. BROWNING / BIOLOGICAL PSYCHIATRY 2020

KNOWABLE MAGAZINE

<https://www.brainfacts.org/thinking-sensing-and-behaving/thinking-and-awareness/2021/psychedelics-open-a-new-window-on-the-mechanisms-of-perception-070821>

Surprising experiences

An **experience** is the unification of multi-modal sensations

An experience is surprising when it is composed of sensations that are prediction errors



At first, an agent holds a trivial causal theory that simply predicts unchanging sensations

Expecting a world that never changes leads quickly to prediction errors

The agent updates its causal theory when too many experiences are surprising

The agent uses its causal theory to predict incoming sensations but also to predict what actions favor good experiences over bad ones

Attention sets the goals

Not all experiences are worthy of being acted on

The agent pays **attention** to the experiences that matter most to its survival, those it feels most intensely

Not only the ones that feel worse but also the ones that feel best

The **confidence** an agent has in its experiences modulates how it pays attention to them

The agent assigns itself the **goals** of impacting experiences it pays attention to

An agent seeks to gain confidence in its experiences before attempting to persist good ones and terminate bad ones

An agent has confidence in experiences that are *not* surprising or, if they are, that are *enduring*

Even if an experience is surprising, the longer it persists the more confident the agent is about it

Prioritizing existential goals

An agent prioritizes goals that can improve its wellbeing



Pursuing exploration goals

When survival is not at stake, an agent can act for the sake of new experiences

It babbles



Implementation

Project challenges

Somehow, the robot must be a **mortal agent**

So it can have *good vs bad* experiences and take *beneficial* actions

Somehow, the robot must **experience** its world, **make sense** of its experiences, and **act** to promote good ones

Somehow, the robot's experiences, sense making and goals must be ***entirely its own***

What of Active Inference?

Active Inference does not tell me how to program a mortal robot that actively learns how to survive

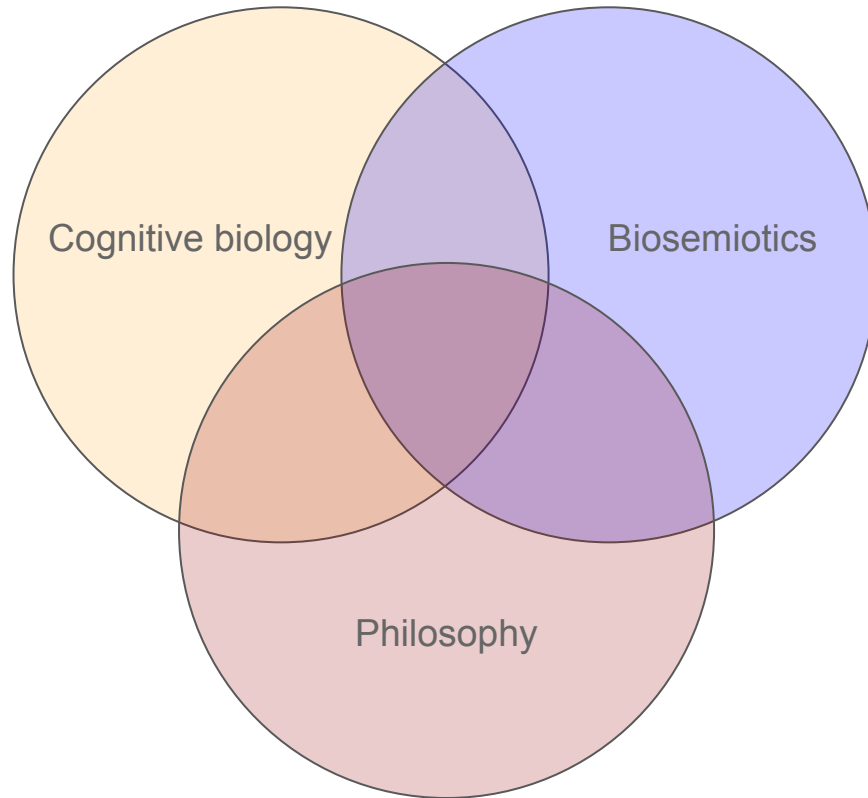
It tells me what information dynamics are required for the robot to do so

Active Inference sets up **requirements**

The observable states of the robot must conform to the *expectations* set forth by the Free Energy Principle

Building a robot's mind

The implementation draws inspiration from...



Ideas from cognitive biology

Organism as mutually sustaining/constraining processes

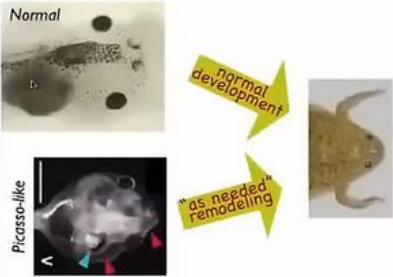
Multiscale competency

Stress as cognitive glue

Problem-solving in morphospace

Cognitive light cones

Intelligent Problem-solving in Morphospace



Normal development

as needed remodeling

Genetics does not specify hardwired rearrangements: it specifies a system that executes a highly flexible program that can recognize unexpected states and take corrective action.

Cannot just follow a rote set of steps. How does it know when it's "right"?

Change bioelectric prepatterning
↓
Craniofacial mispatterning
↓
Metamorphosis
↓
Morphometric analysis and modeling
reveals: **faces fix themselves!!**



dorsal profile



A B



Institute for
Computationally Designed
Organisms



Michael Levin

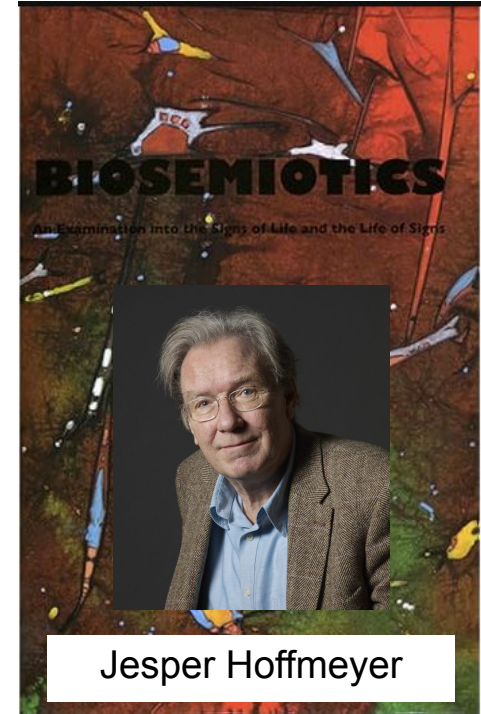
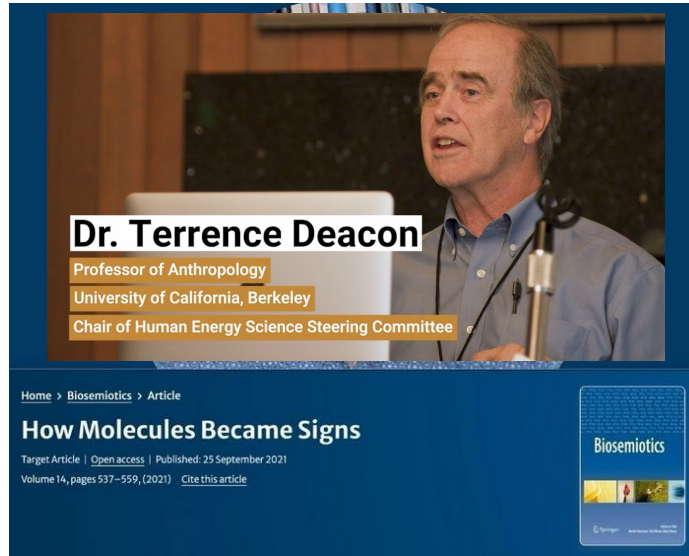
Ideas from biosemiotics

Umwelts

Aboutness

Signal processes

Sense making



Ideas from philosophy

Kant's Synthetic Unity of
Apperception

Google DeepMind's
Apperception Engine



Richard Evans

Deep Mind

The Apperception Engine

Talk outline: Imagine a machine, equipped with various sensors, receiving a stream of sensory information. Somehow, it must make sense of that sensory stream. But what, exactly, does "making sense" involve, and how can it be implemented in a machine? I argue that making sense of a sensory stream involves constructing objects that persist over time, with properties that change over time according to intelligible universal laws (represented as logic programs). Thus, making sense is a form of program synthesis - but it is unsupervised program synthesis from raw unstructured input. I will describe our system, the Apperception Engine, that was designed to make sense of raw sensory input by constructing programs that explain that input. I will show various examples of our system in action. This talk will draw on two of our recent papers published in the Artificial Intelligence Journal: "Making Sense of Sensory Input", and the sequel "Making Sense of Raw Input".

Bio: Richard Evans is a Staff Research Scientist at DeepMind and an Honorary Senior Research Fellow at Imperial College. He is particularly interested in program synthesis, neuro-symbolic systems, and the philosophy of Immanuel Kant.

The **software** is built using Elixir, Prolog and CHR

Elixir is a process-oriented and functional programming language

Prolog is a logic programming language

CHR (Constraint Handling Rules) is a constraint solving extension to Prolog

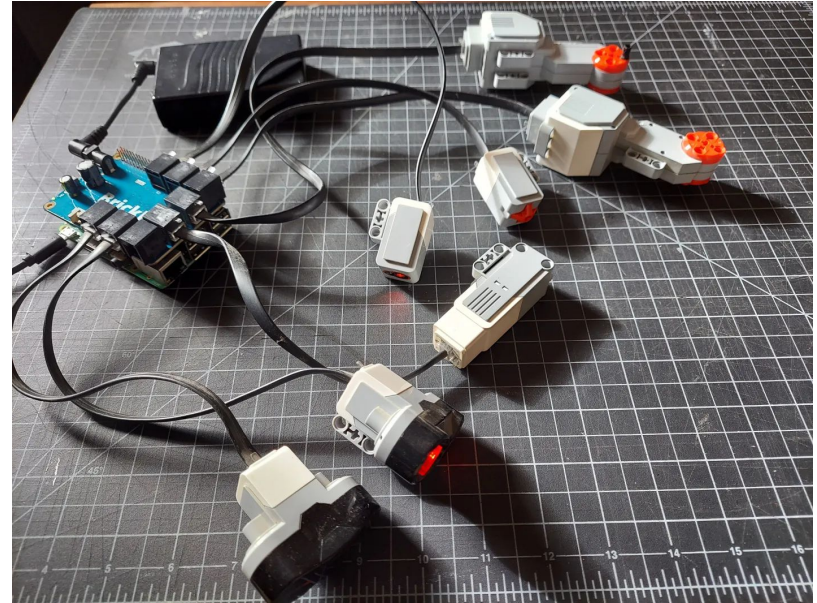
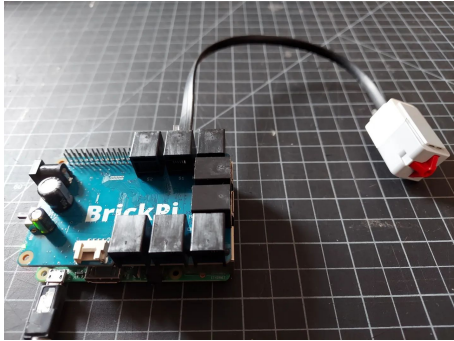
The **hardware** is the robot's devices plus a high-end server

Robot body hardware

Lego EV3 sensors and motors

A Raspberry Pi

A Dexter Industries BrickPi board to interface with the EV3 sensors and motors



Robot mind hardware

On the backend, a 128-core server
does the compute-intensive
cognitive processing



Writing mortal code

Can a robot be made mortal?

The robot's physical structure is immortal but the *processes* animating it come in and out of existence

A robot's *mind* can be made mortal

The robot grows a *mortal mind* to animate an *immortal body* so it can survivably explore/exploit its world

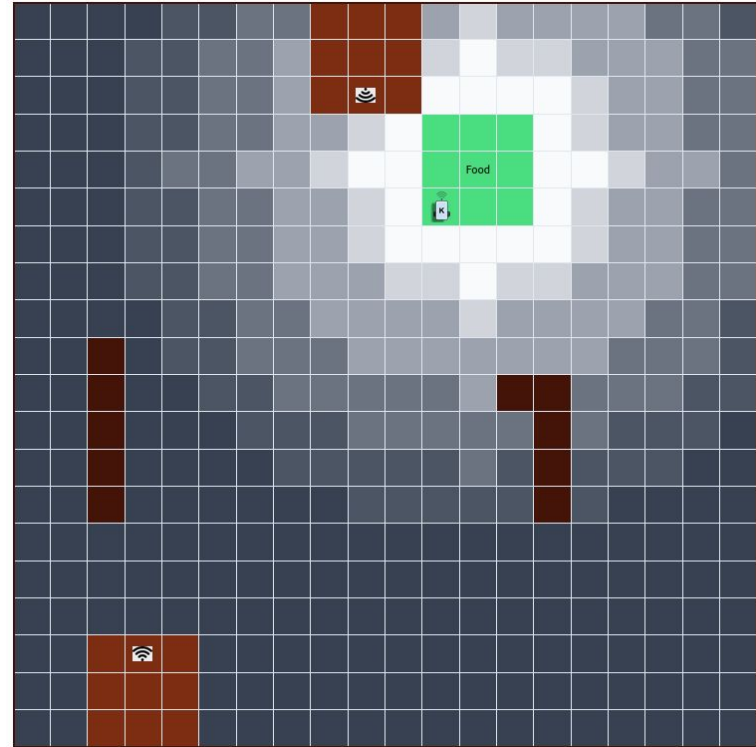
The robot starts with a minimal mind and a reserve of wellbeing

It must learn to maintain sufficient wellbeing or else its mind dies

Virtual grid world

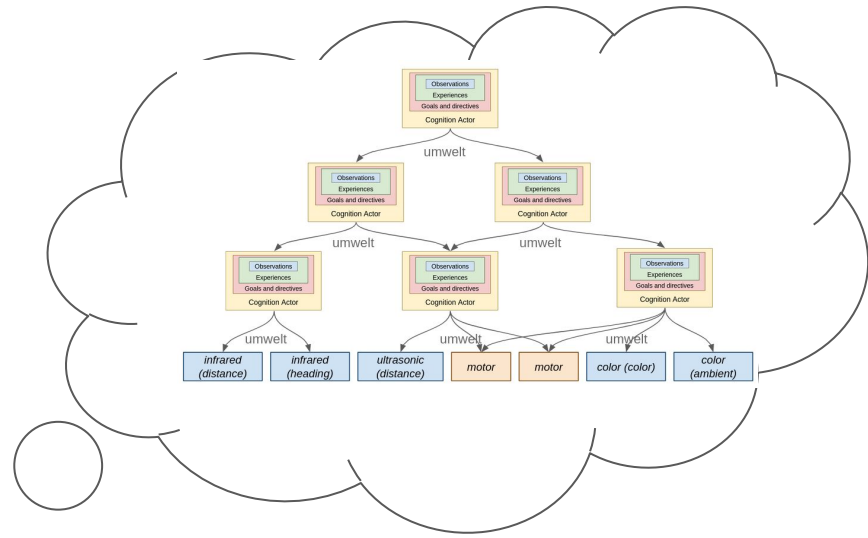
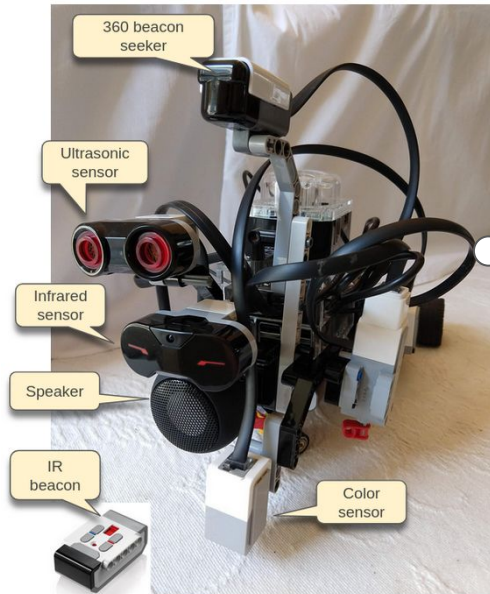
A virtual grid world provides a virtual robot with an environment that can both nourish (food) and damage it (obstacles, poison)

A physical environment will be built that recreates the conditions of the virtual world but with the messiness of the analog world



Growing the robot's mind

The robot's mind is an evolving collective of long-lived processes

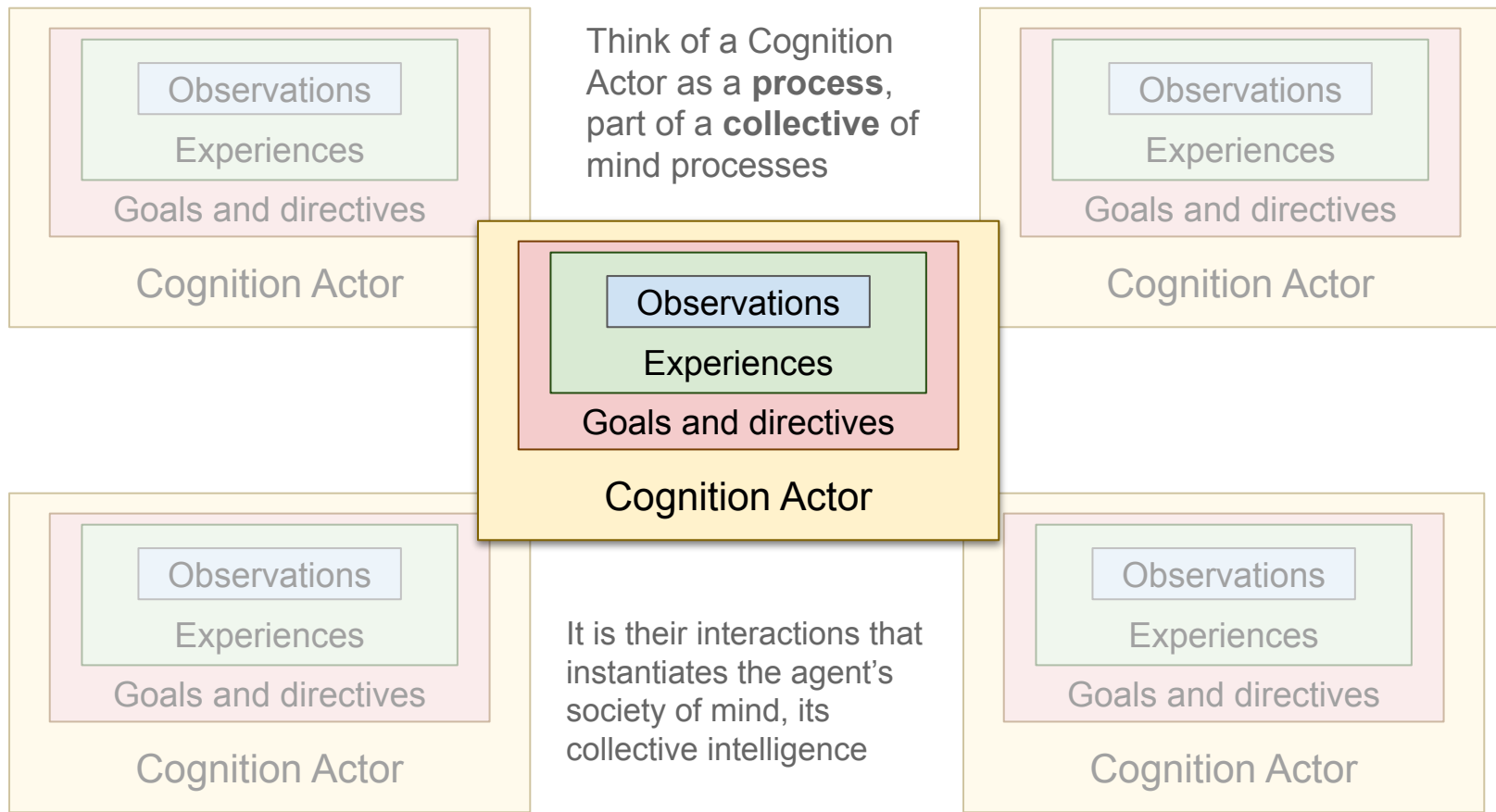


All intelligence is collective intelligence

The robot is not a monolithic agent

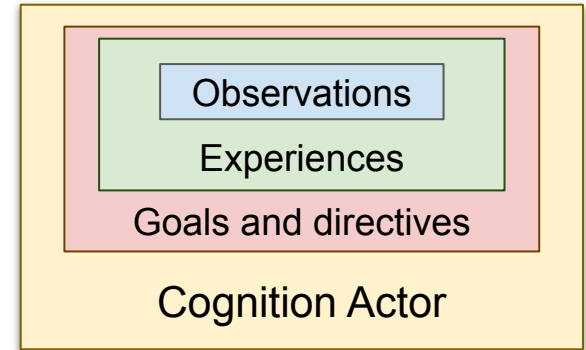
It contains multitudes

The robot's mind is implemented as a dynamic **collective**
of **cognition actors (CA's)**



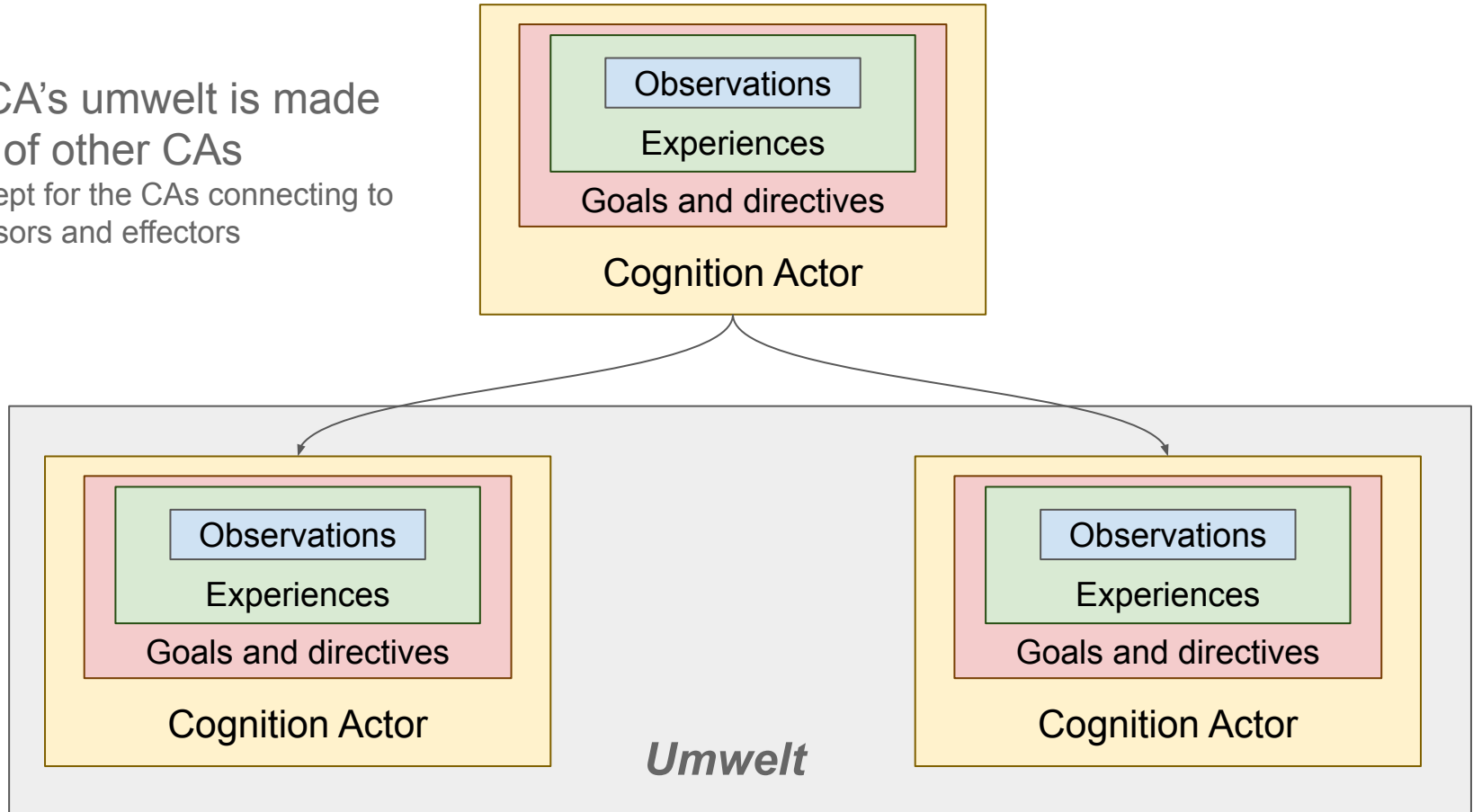
Each cognition actor (CA) is implemented as a *long-lived, stateful process* that

- makes **observations** (predictions + prediction errors)
- **makes sense** of these observations
- **unifies** observations into **experiences**
- assigns **value** to them
- **acts** to have better experiences

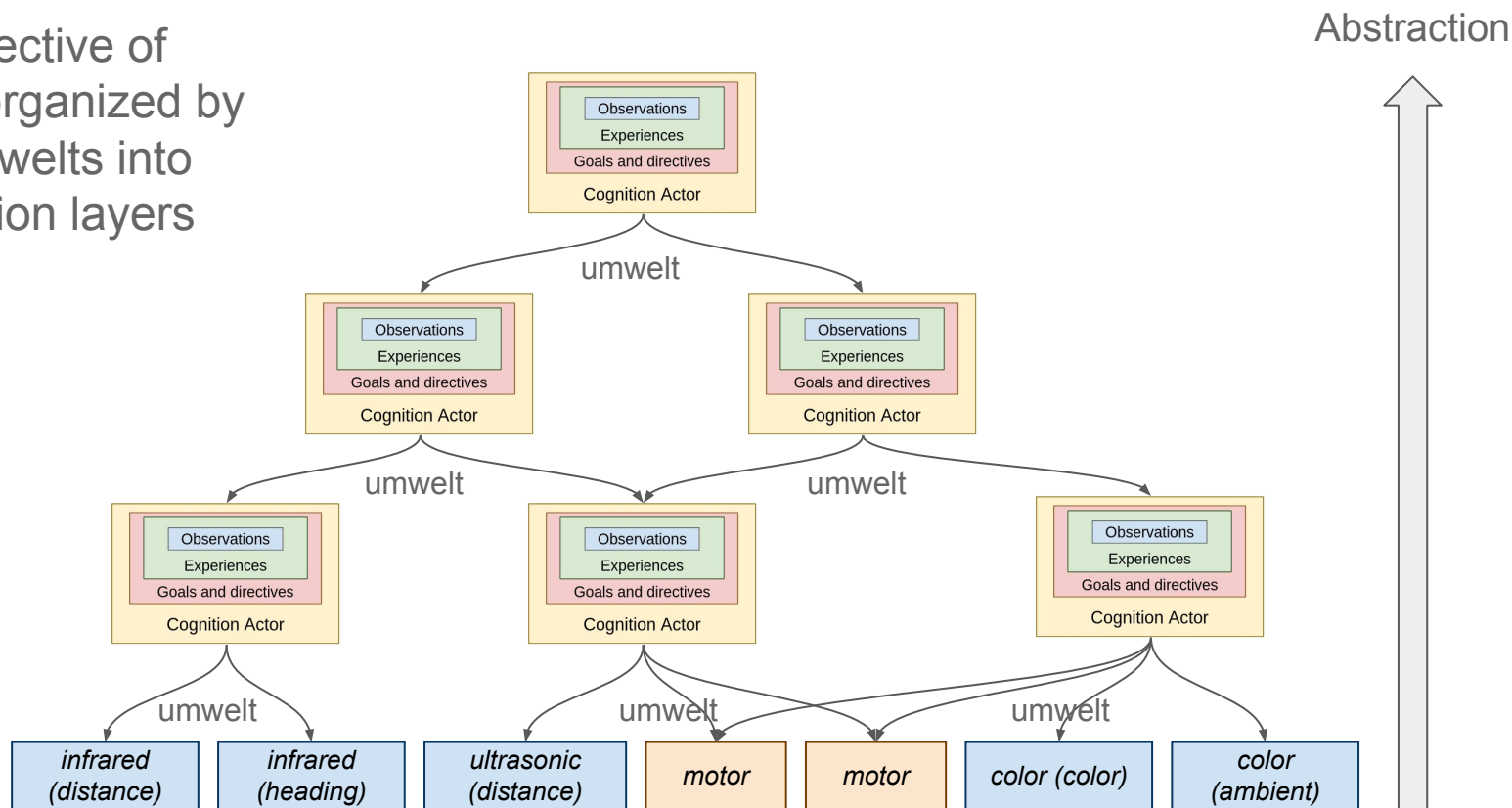


Each CA has its own **umwelt** (its self-centered world) that it observes, makes sense of and acts on

A CA's umwelt is made
up of other CAs
except for the CAs connecting to
sensors and effectors



The collective of CAs is organized by their umwelts into abstraction layers



The bottom layer is populated a priori by **static CAs**

They are static only in the sense that they are permanent

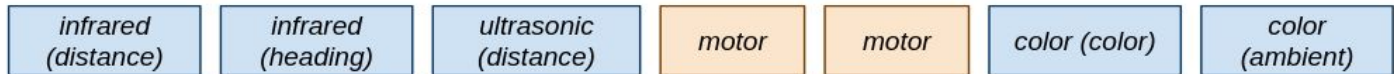
Static CAs interface directly with the robot's sensors (sensor CAs) and motors (effector CAs)

Their umwelts is the robot's sensed world

Static CAs connect the robot's mind with the body's sensors and motors



Static CAs



All other layers of CAs come in and out of existence
during the lifetime of the robot's mind

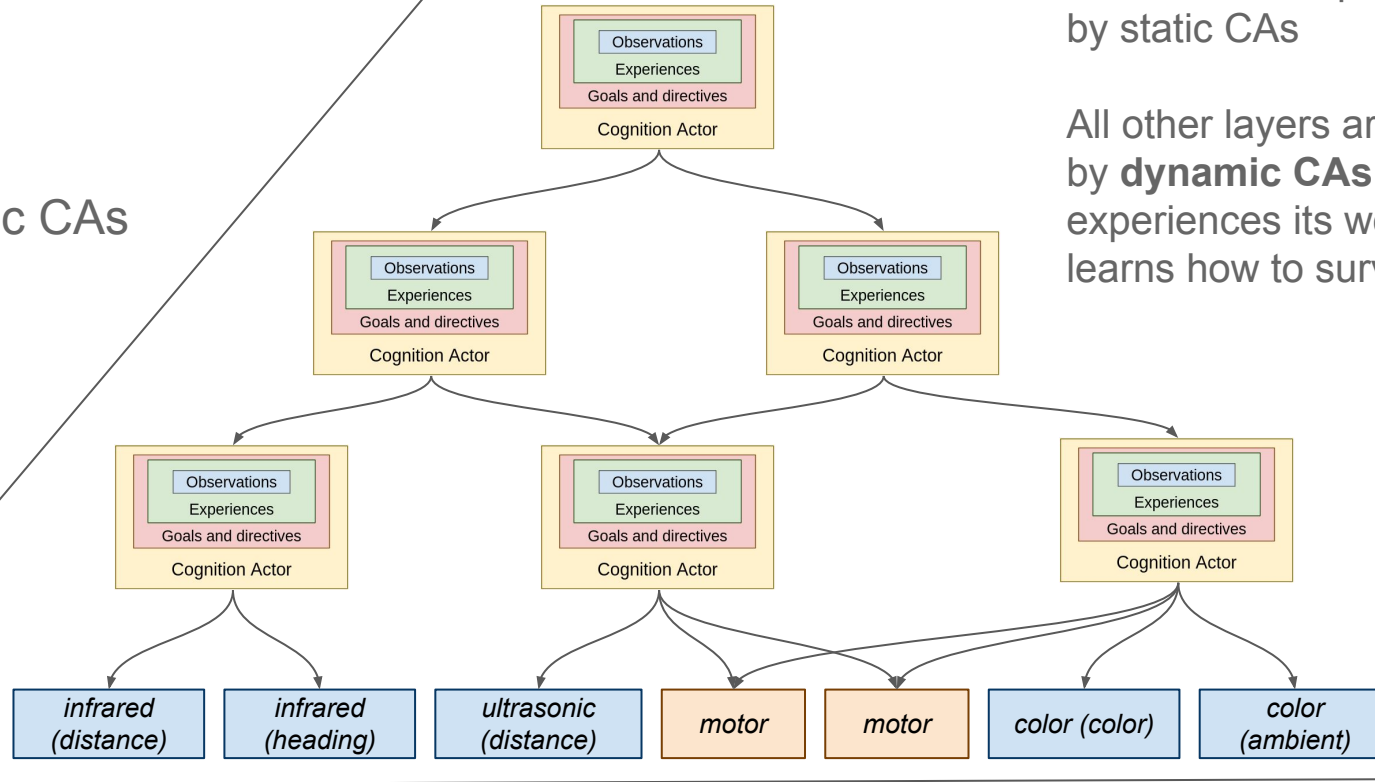
They capture what the robot learns from striving to survive

Survival of a collective of processes

The *a priori* bottom layer of static CAs is not enough by itself for the robot to find nourishment and avoid hazards

Additional layers of CAs are needed to create the “big picture” experiences and affordances needed for competency and survival

Dynamic CAs



The first row is pre-populated by static CAs

All other layers are populated by **dynamic CAs** as the robot experiences its world and learns how to survive

Static CAs

A society of mind grows, shrinks and reshapes itself by adding and removing dynamic CAs

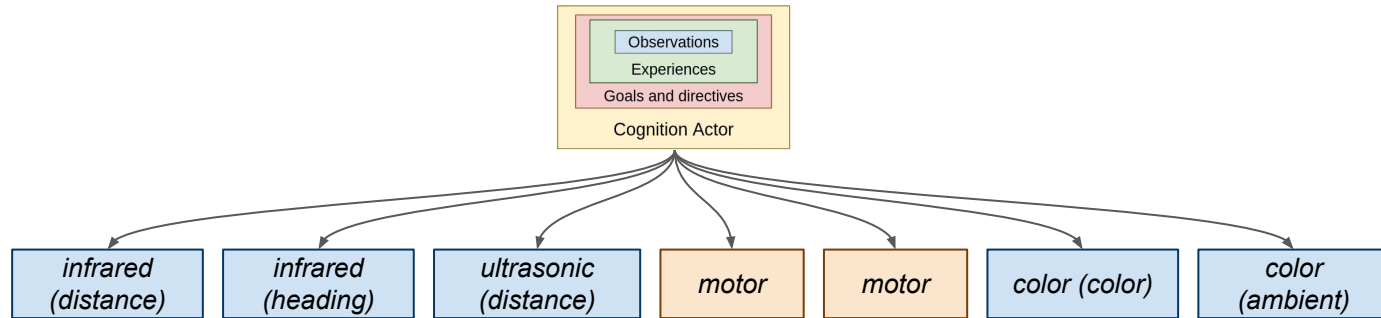
A society of mind shapes itself to capture what it learns as it strives to survive

It searches for a competent self-organization in the space of all possible “shapes” it could take (its morphospace)

To help the search converge to a competent organization,
the collective applies principles from developmental
biology

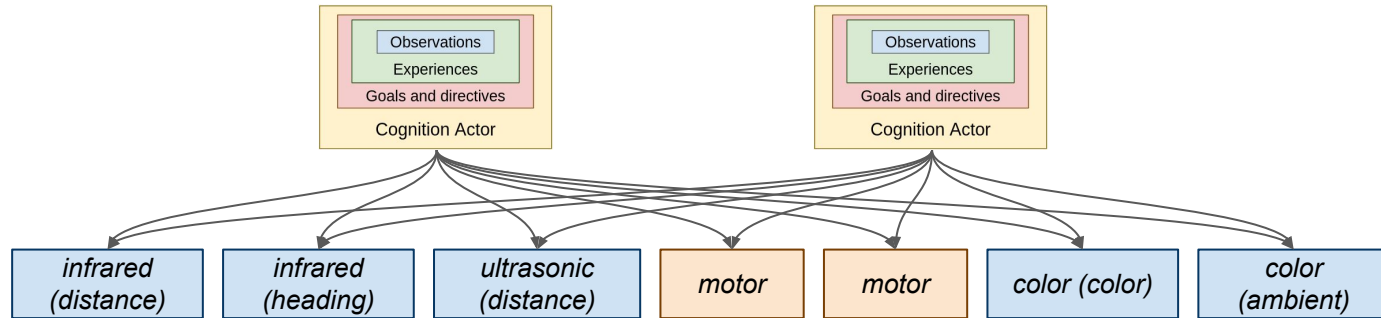
- **Parts from wholes, not wholes from parts:** Grow by replication, division and differentiation, not by assembly
- **Use it or lose it:** Grow to over-abundance then trim to essential
- **Stress as cognitive glue:** The stress felt by parts at risk propagates and motivates global change
- **Stable wholes, transient parts:** The simpler the parts, the faster they are replaced

The first dynamic CA to occupy a layer grabs the whole layer below as its umwelt



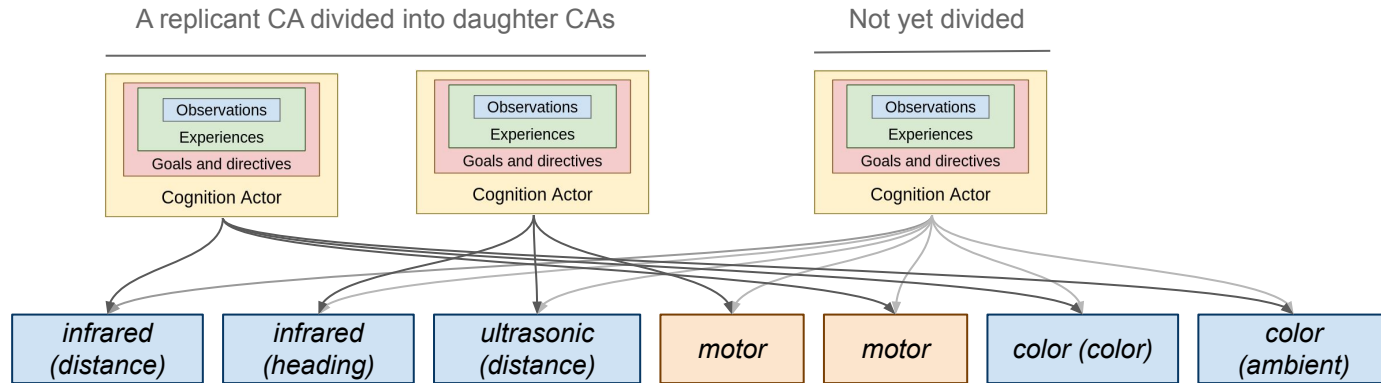
The dynamic CA then *replicates* until it saturates its own layer with replicants

Replicants *differentiate* their internal states over time because life cycles are not deterministic



After saturating their layer, replicant dynamic CAs begin to divide into daughter CAs

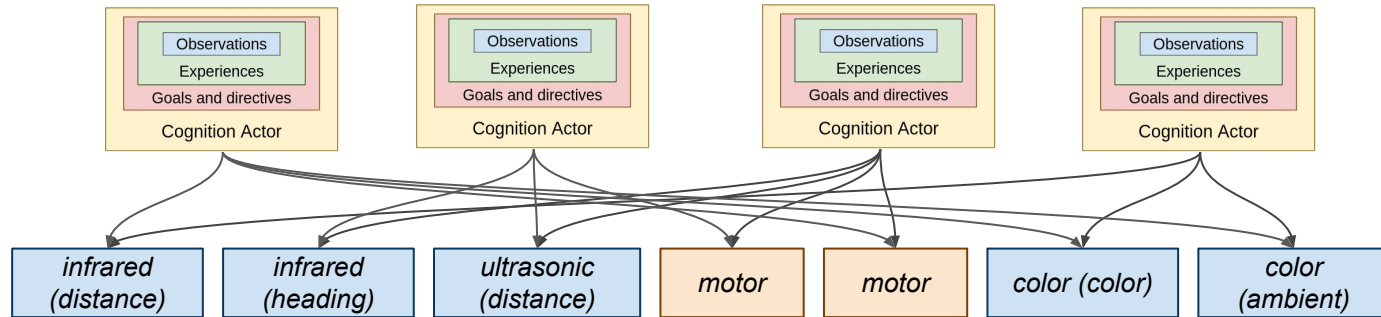
Each daughter CA gets a random half of its mother's umwelt



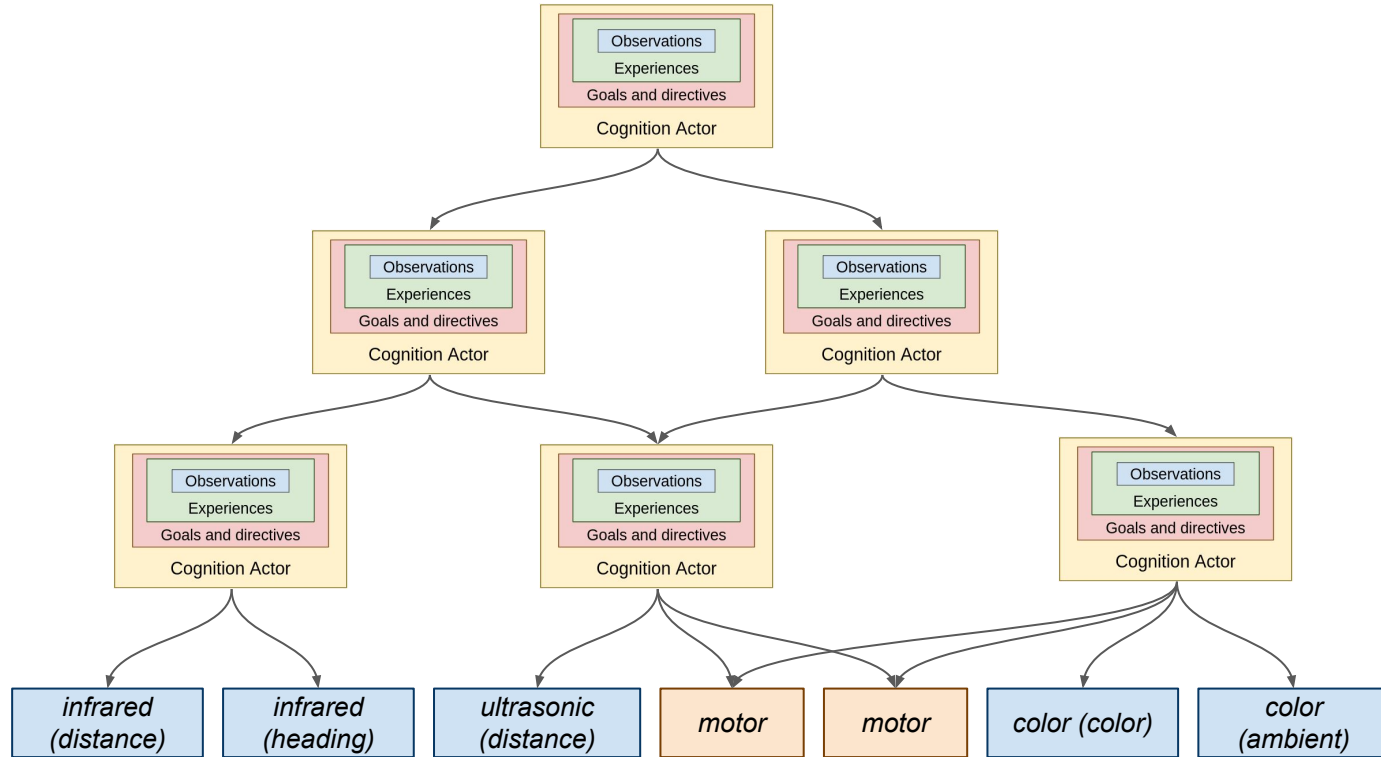
Dynamic CAs keep dividing until then can no longer do so

At that point, the layer is full and a new layer can be started

Full layer



Over time dynamic CAs grow into an abundant abstraction hierarchy

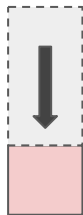


A dynamic CA will remove itself if it persistently serves no purpose

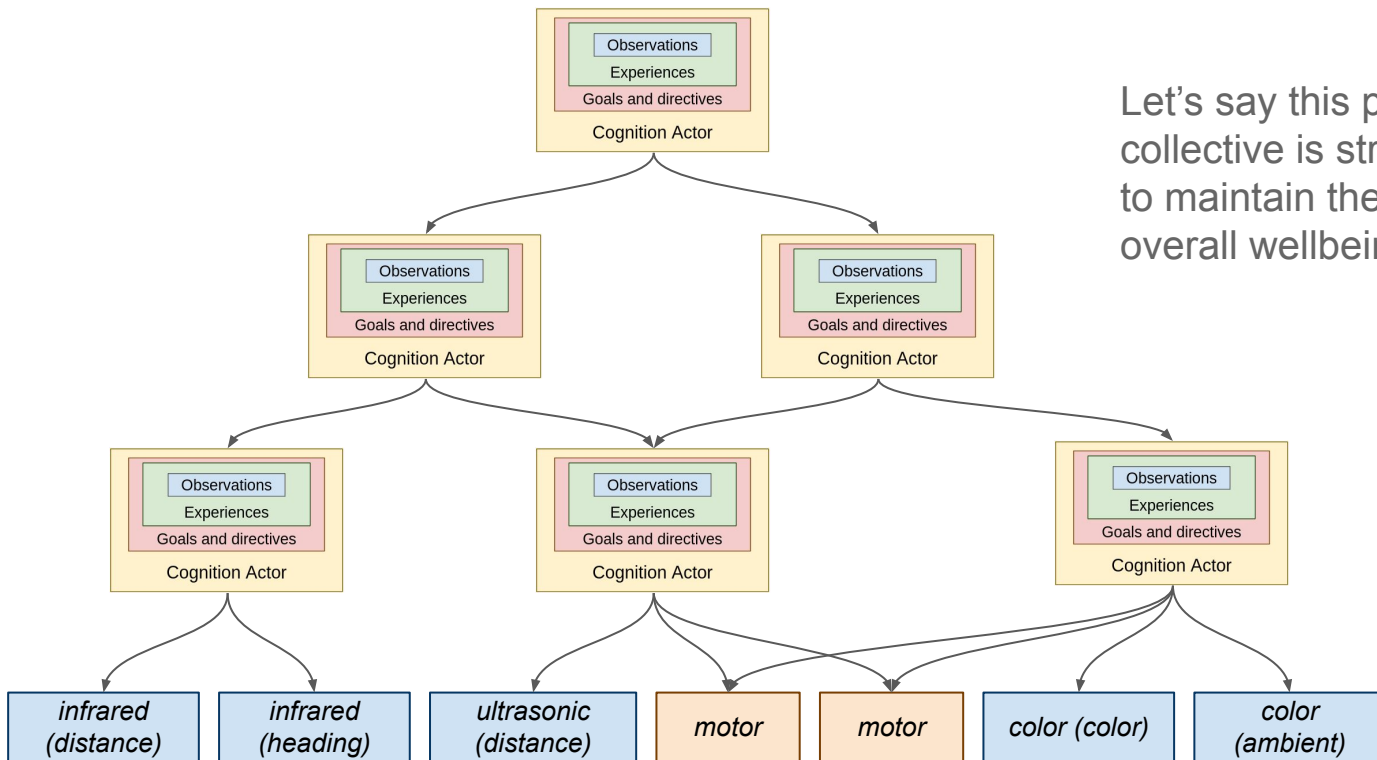
This removes useless CAs and can make room for other CAs to come into existence and perhaps do better

This adding and removing of CAs and layers of CAs explores a space of possible organizations of the mind

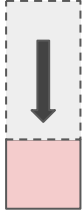
Searching for a **competent** collective



Wellbeing



Let's say this particular collective is struggling to maintain the robot's overall wellbeing

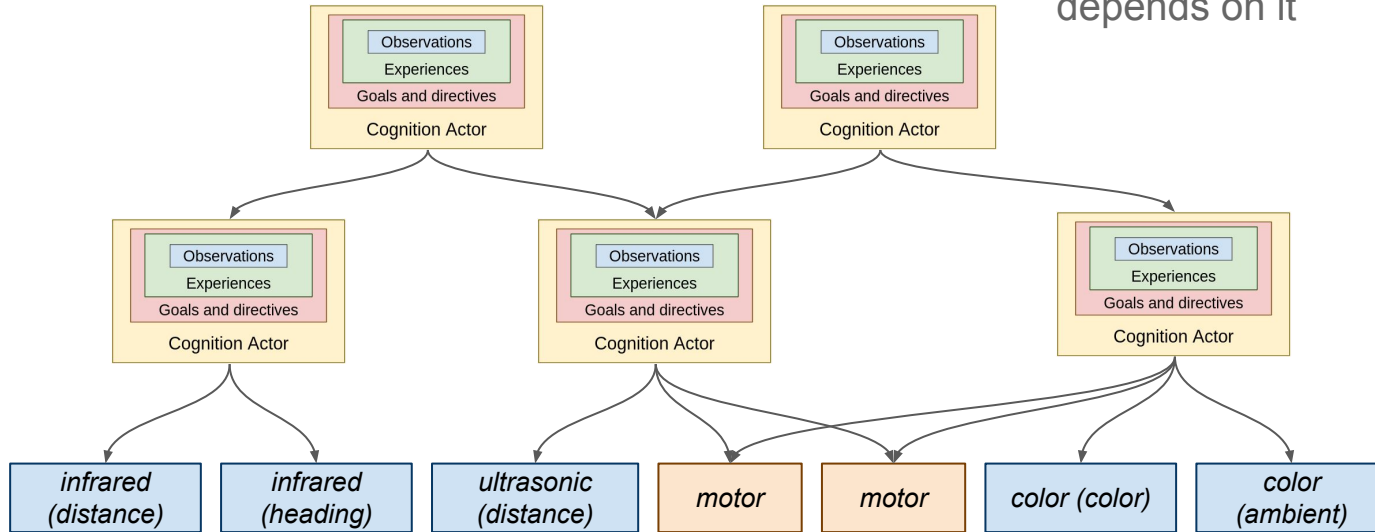


Searching for a **competent** collective

Wellbeing

The top CA removes itself

It can because no other CA depends on it

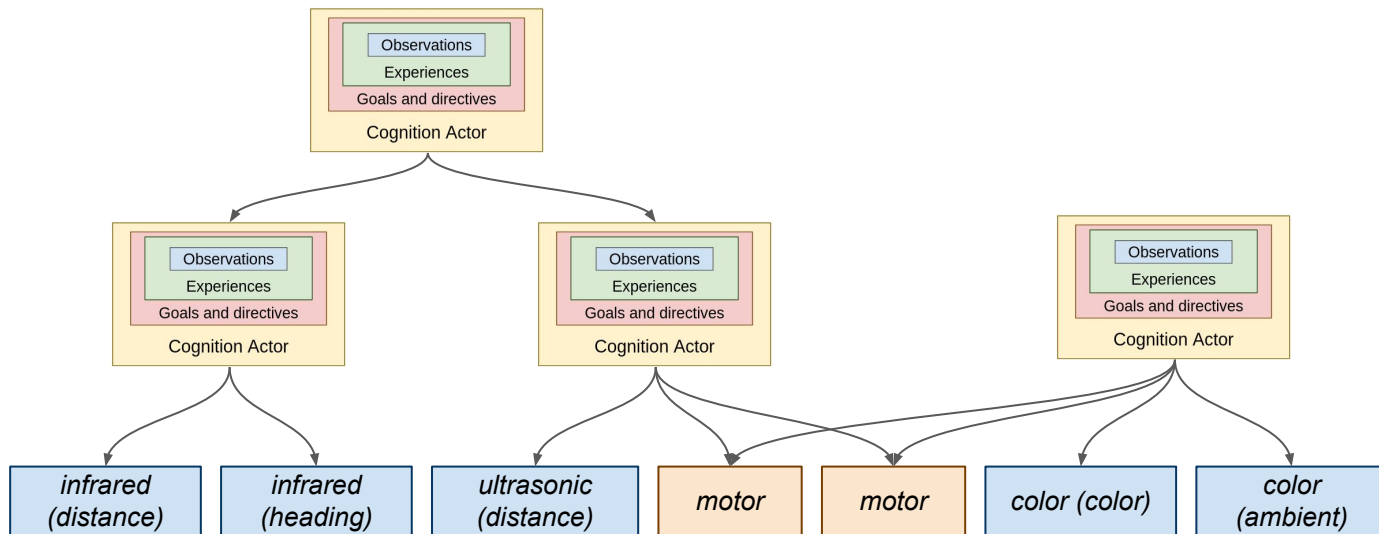


Searching for a **competent** collective

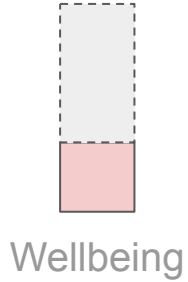


Wellbeing

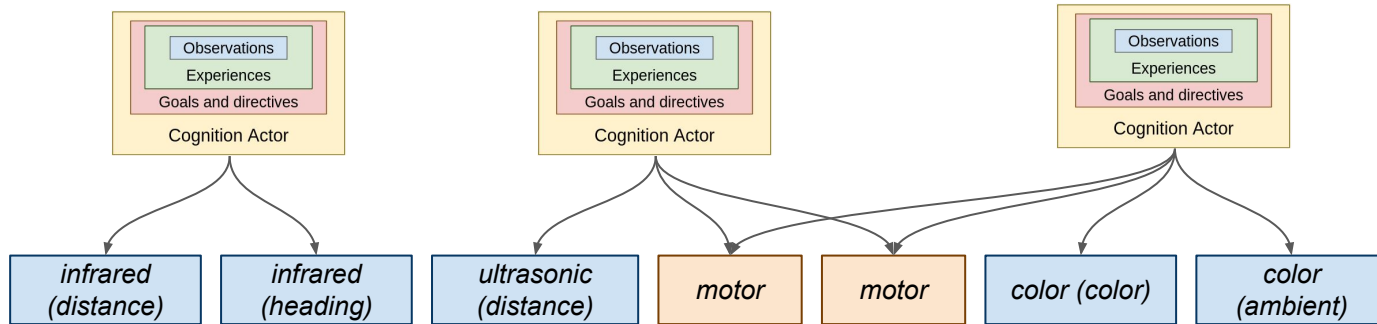
This allows for other underperforming CAs to also remove themselves



Searching for a **competent** collective



And so on, reducing the “energy cost” and thus the loss of fullness by the collective

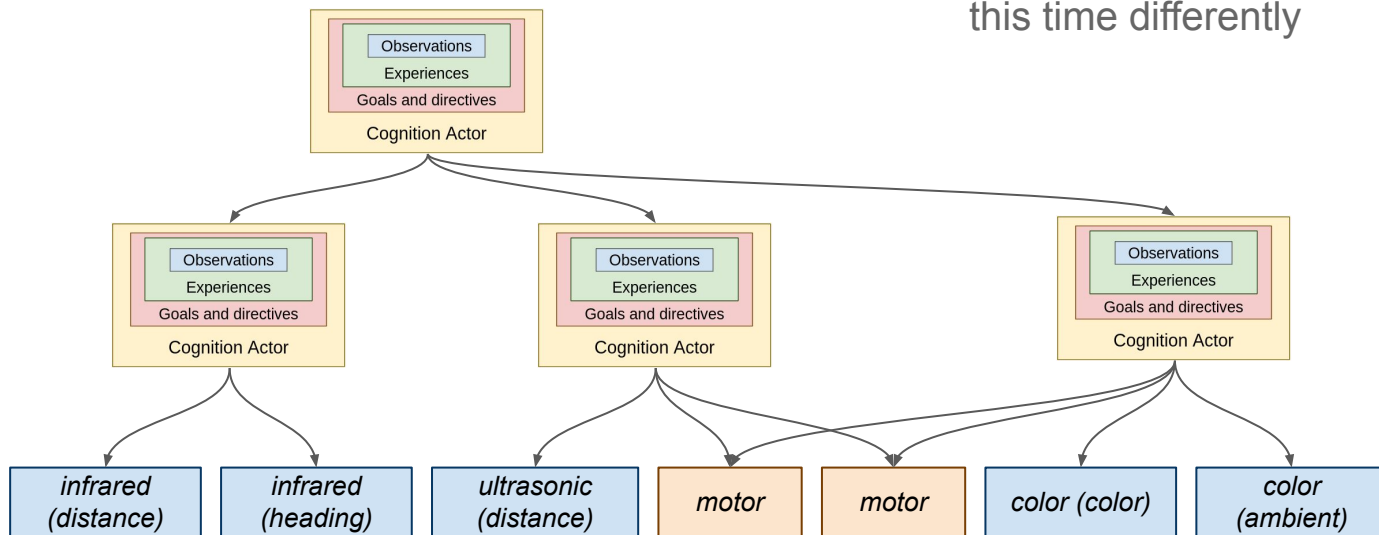


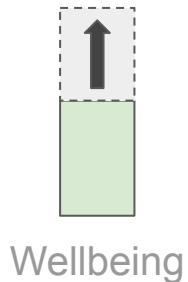


Wellbeing

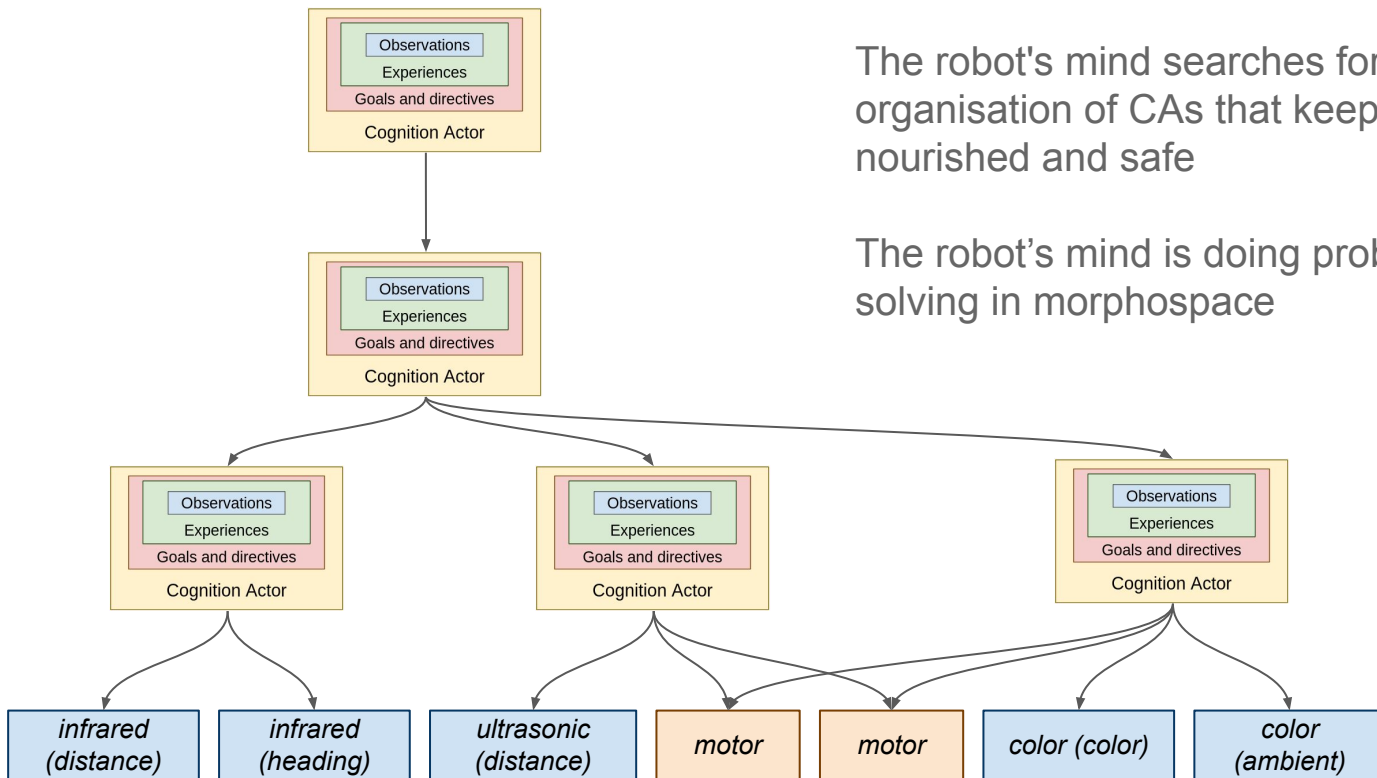
Searching for a **competent** collective

Having a lower “energy draw”, the collective can now afford to regrow itself, this time differently





Searching for a **competent** collective

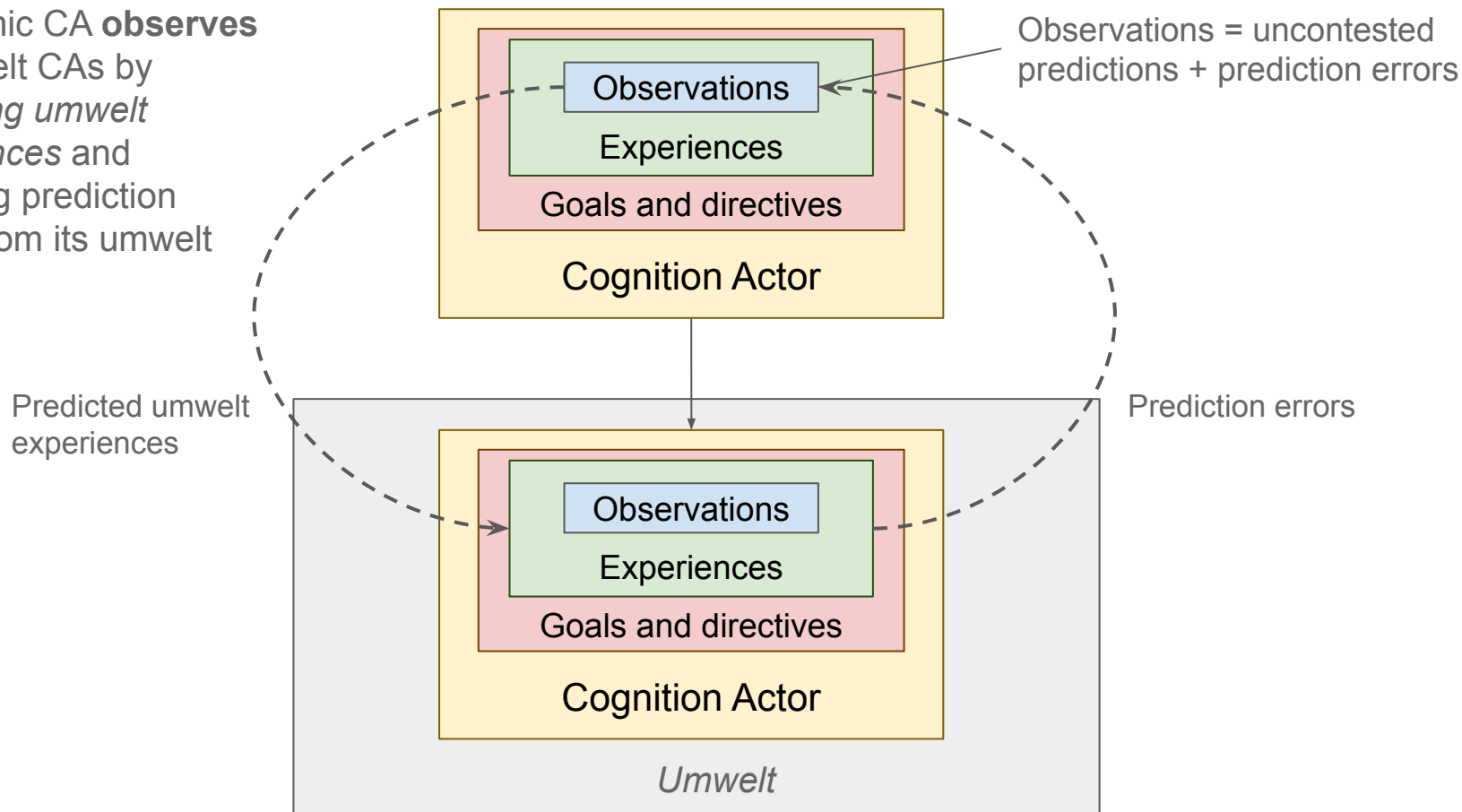


The robot's mind searches for an organisation of CAs that keeps it nourished and safe

The robot's mind is doing problem solving in morphospace

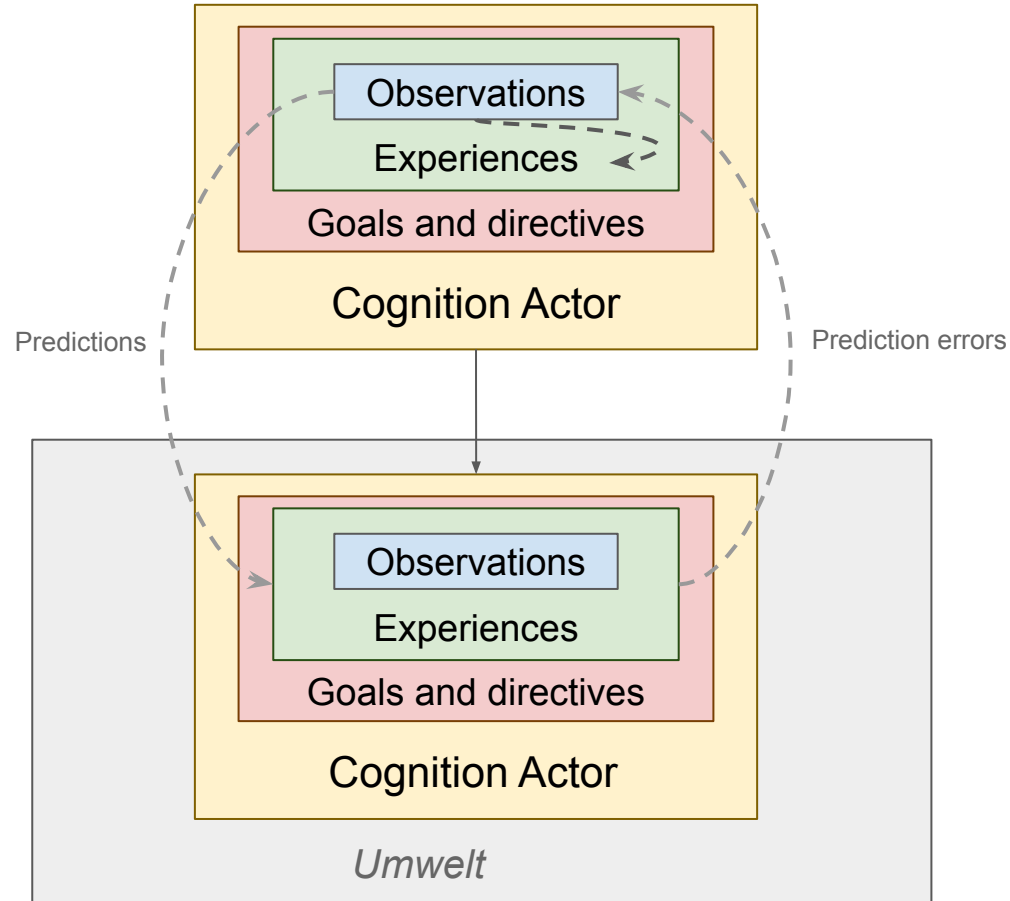
Observing and experiencing

A dynamic CA **observes**
its umwelt CAs by
predicting umwelt
experiences and
receiving prediction
errors from its umwelt



A dynamic CA *unifies* its past and present observations into **experiences**

For example, having observed in its umwelt that distances keep shrinking, a dynamic CA would experience "approaching"



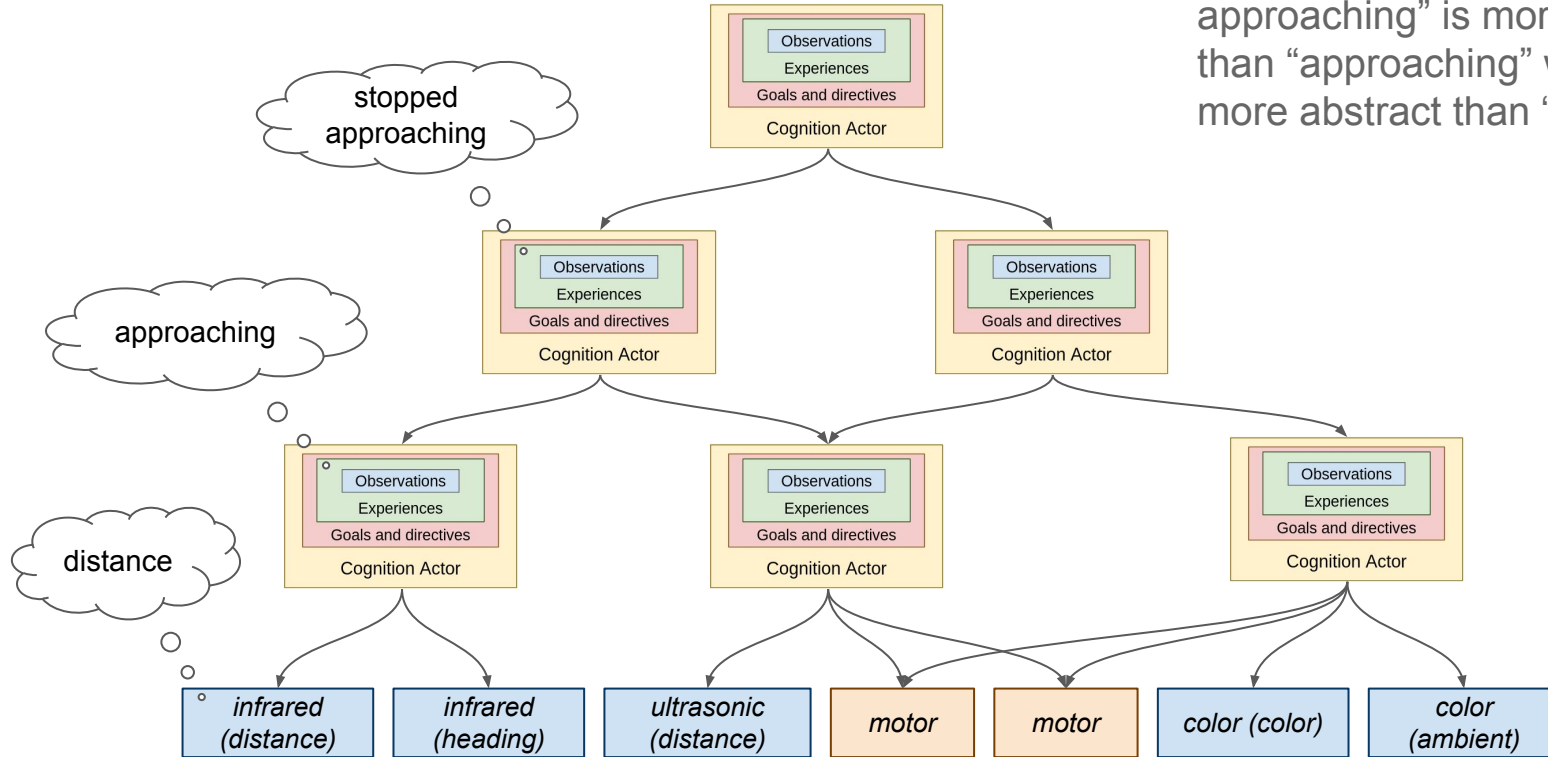
Sensor CAs (static CAs in the bottom layer) have simple experiences from observing distance, color, luminance etc.

The observations of sensor readings are their experiences

While a static CA observes and experiences the readings of its associated sensor, a dynamic CA observes and unifies *the experiences* of its umwelt CAs

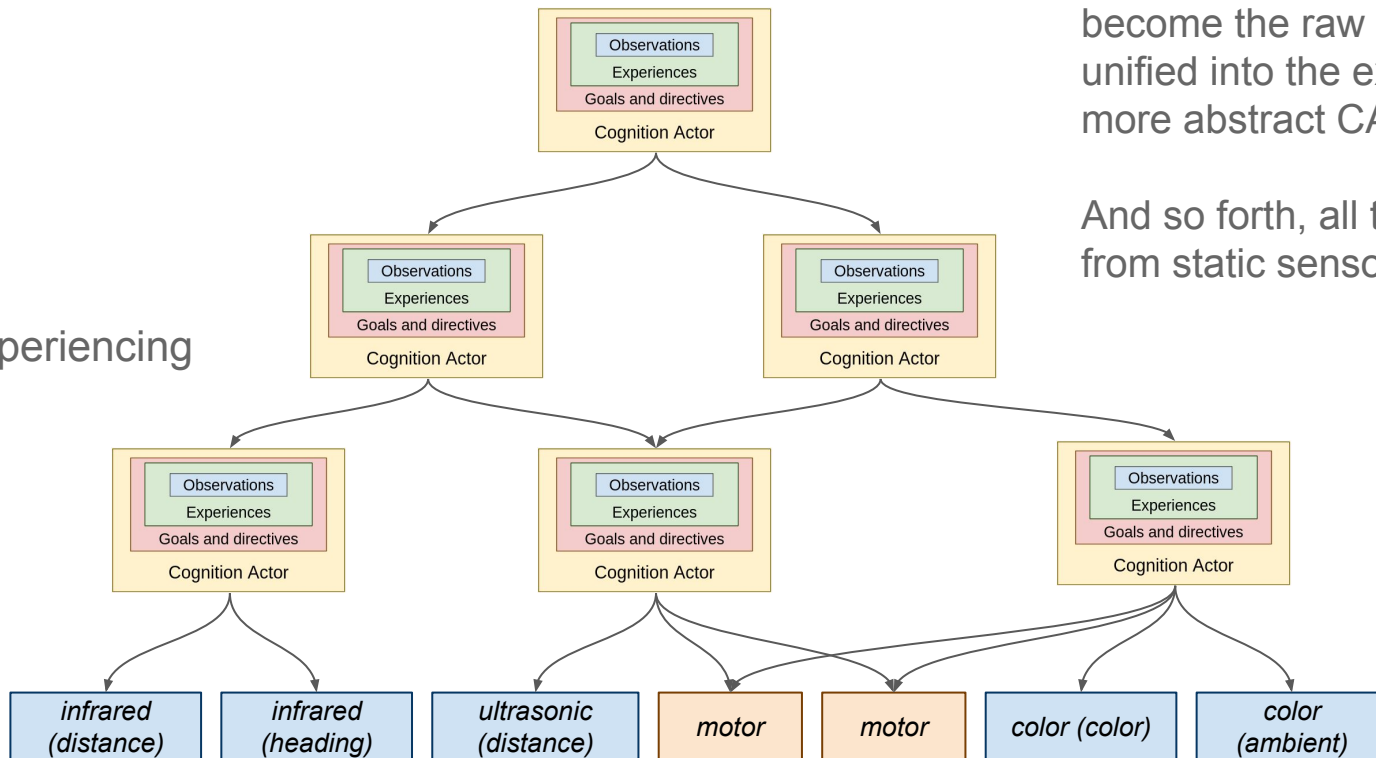
Higher up dynamic CAs have more abstract experiences
than lower down CAs

For example, “stopped approaching” is more abstract than “approaching” which is more abstract than “distance”





Experiencing



The experiences of a CA become the raw material to be unified into the experiences of more abstract CAs

And so forth, all the way up from static sensor CAs

Experiences are constructed and named by dynamic CAs,
not by the programmer

They need not make sense to an external observer

Feelings

The CAs composing the robot's mind feel the fluctuations in their individual wellbeing

Everything a CA does increases or decreases its feeling of wellbeing

Any internal or external work reduces fullness

Spending time over a food patch increases fullness

Hitting an obstacle reduces integrity

Participating in an umwelt increase engagement, etc.

Wellbeing measures are not separate

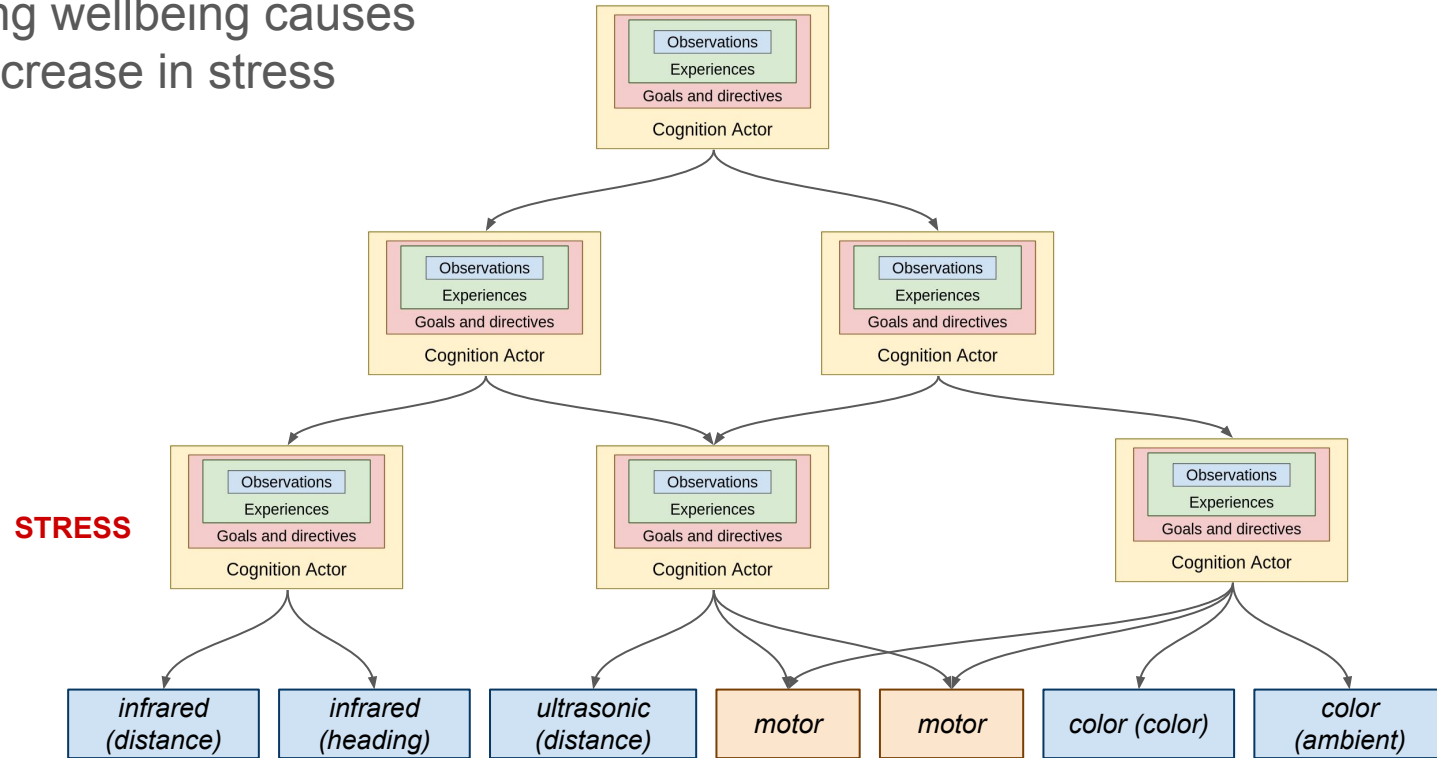
Activating motors is work that causes loss of fullness and integrity (wear and tear), but also increases engagement

Lost integrity is regained over time (think healing) by draining fullness, etc.

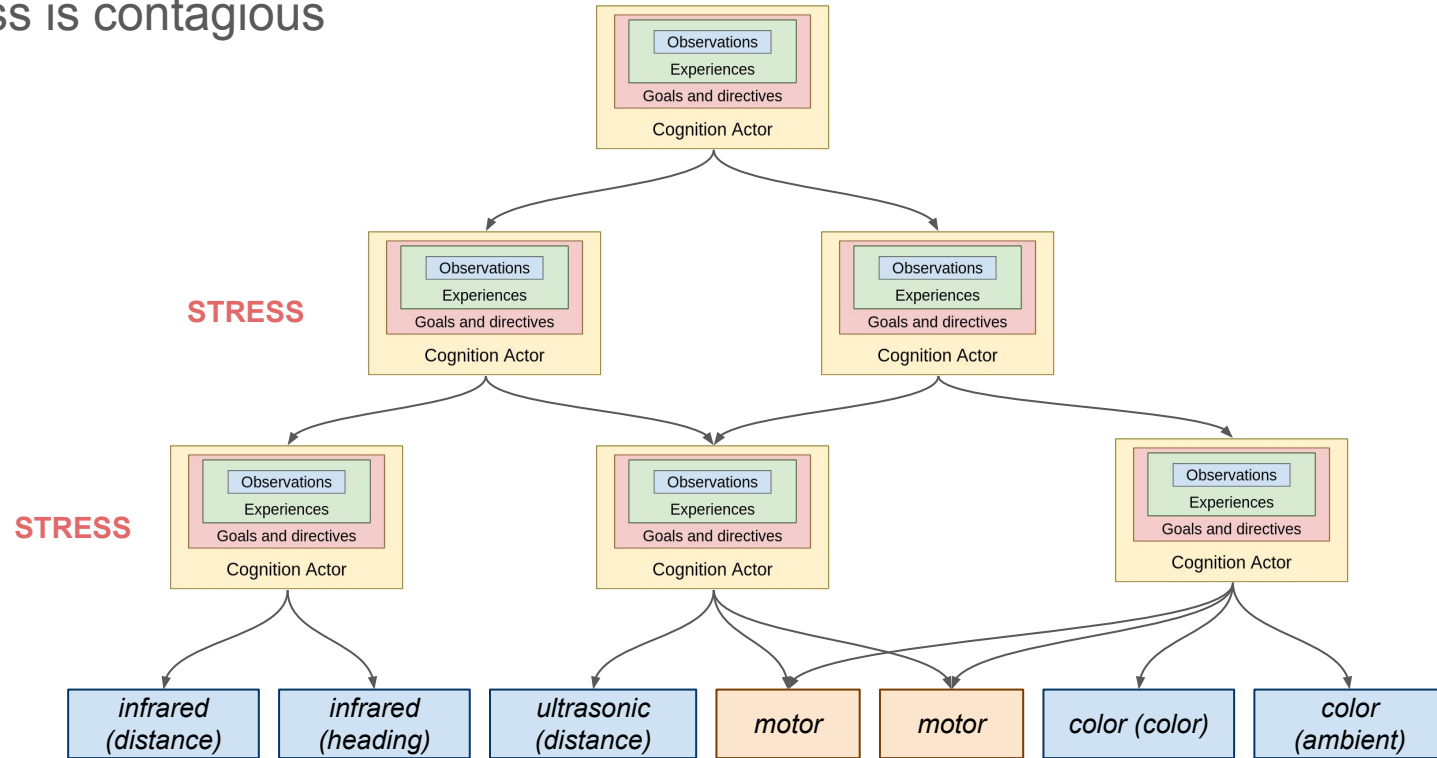
A CA's fluctuating wellbeing determines whether its *concurrent* experiences feel good or bad

The loss or gain in wellbeing of a CA diffuses throughout the collective as “stress” (increasing or decreasing)

Losing wellbeing causes an increase in stress



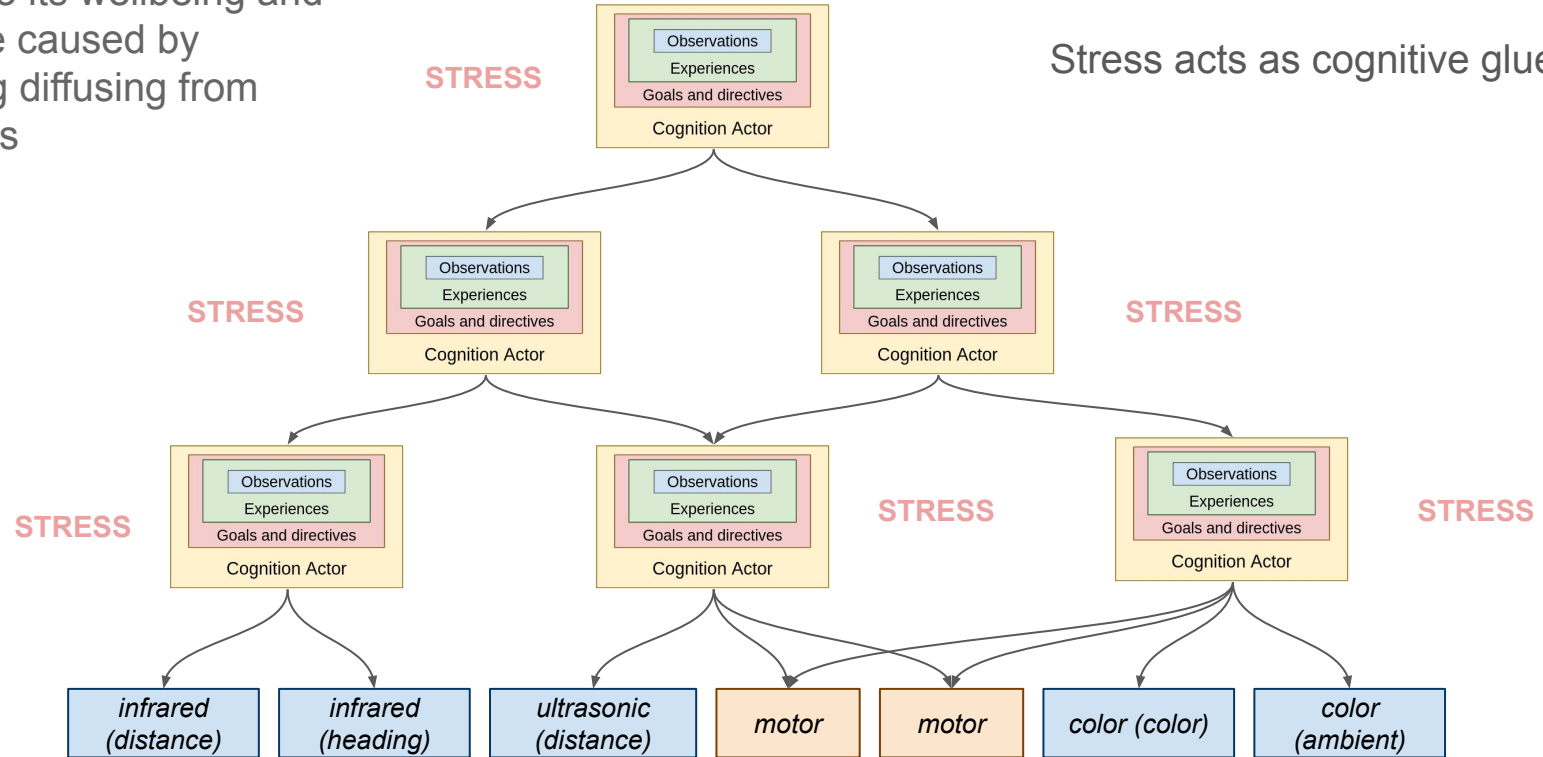
Stress is contagious



A CA can not distinguish between a self-induced change to its wellbeing and a change caused by wellbeing diffusing from other CAs

This is how a CA's problem becomes every CA's problem

Stress acts as cognitive glue



Stress alters the behavior of CAs

A “stressed” CA is more likely to try something different from what has worked in the past

A “comfortable” CA is more likely to settle into *habits*

Acting

A dynamic CA acts to confirm its experiences, terminate its worse ones and persist the better ones

A dynamic CA seeks to confirm, persist or terminate an experience it pays attention to

To do so, it must first identify the *observed* umwelt experiences that it *unified* into the experience it now seeks to impact

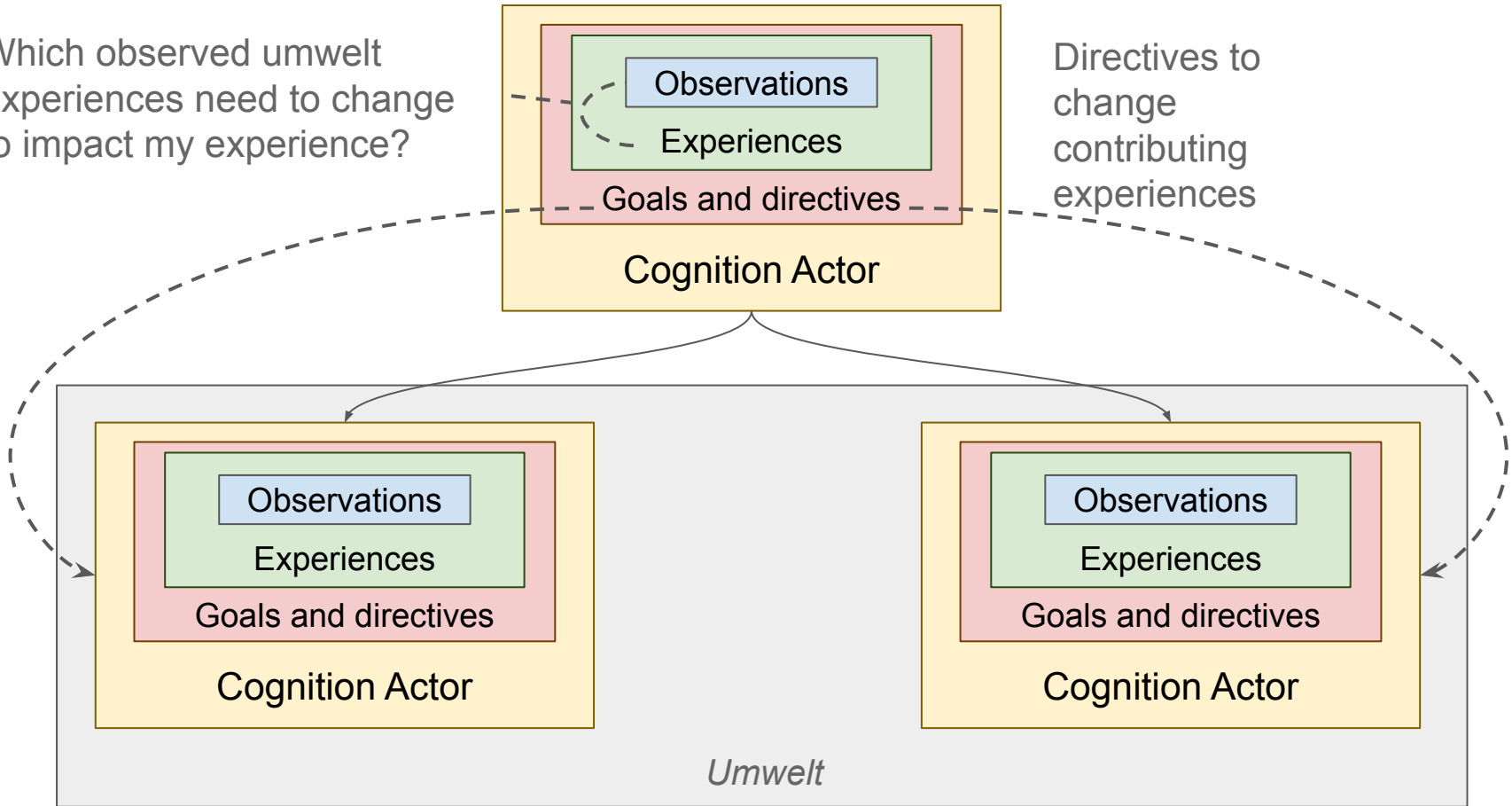
The dynamic CA builds a **plan** to impact observed umwelt experiences that were unified into its own experience

A plan is a set of **directives** it sends to its umwelt

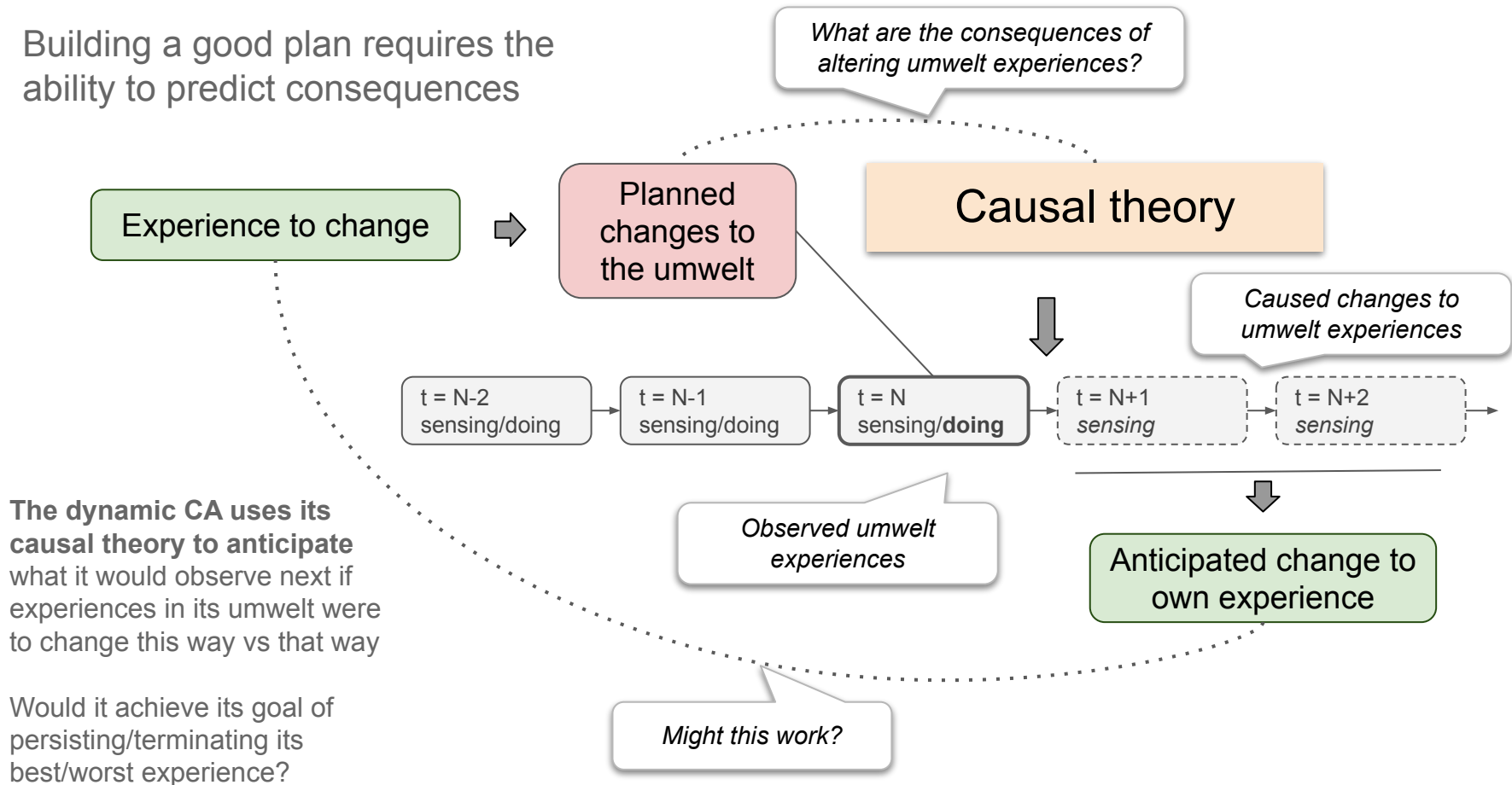
Each individual directive is a **prioritized goal** to initiate, persist or terminate a contributing umwelt experience

Which observed umwelt experiences need to change to impact my experience?

Directives to change contributing experiences



Building a good plan requires the ability to predict consequences



When a dynamic CA receives directives (goals to accomplish) from a parent, it decides which ones, if any, it can accomplish and whether they supercede its own self-assigned goals

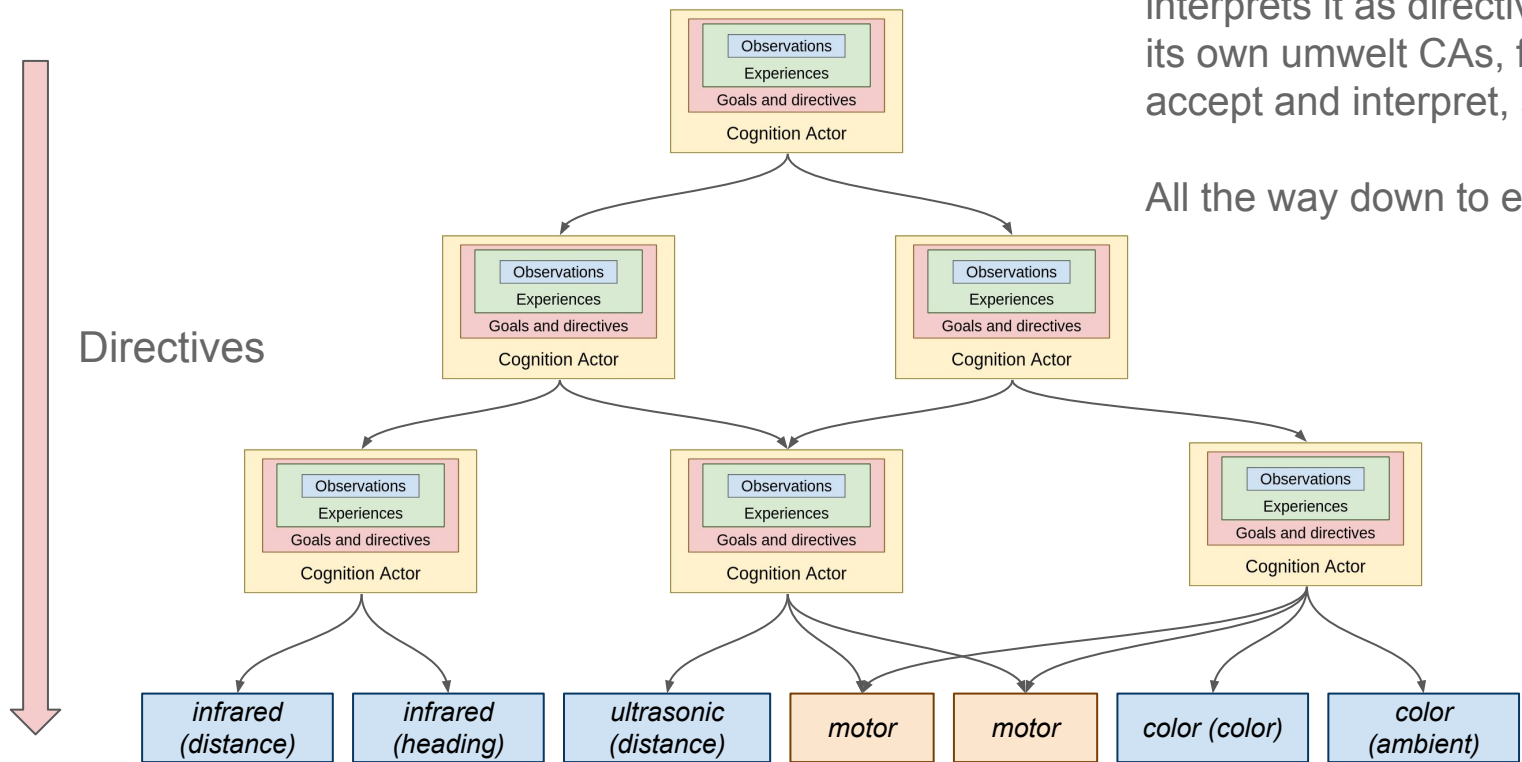
A CA does not micromanage its umwelt

Umwelt CAs *interpret* the directives they receive and accept by making their own plans

The collective of CAs is a multi-scale competency

For each goal it takes on, a CA interprets it as directives it sends to its own umwelt CAs, for them to accept and interpret, and so on

All the way down to effector CAs



A dynamic CA keeps track of plans sent and accepted,
and the goals they were meant to achieve

Plans that appear a posteriori to have achieved their goal
are remembered as **affordances** for later reuse

Affordances become *habits*

CAs higher up discover abstract affordances such as "go forward" or "go up the luminosity gradient"



Affordances are discovered and named by dynamic CAs,
not by the programmer

They need not make sense to an external observer

Causal theories as logic programs

A CA's observes its umwelt one **time frame** after another

Observations made within a time frame are considered synchronous

The more abstract the CA, the longer its time frame for observing, experiencing and acting

A more abstract CA has a bigger “cognitive light cone”

A dynamic CA constructs a **causal theory** to make sense of what it observes within time frames

It is a form of **induction**, that is, the production of general rules from specific examples

A causal theory is a set of **if-then rules** about

- what observations ought to co-occur in a time frame
- what observations can not
- what observations follow previous observations

For example

- *If the distance is getting shorter then nothing is being touched*
- *The distance can not be getting both smaller and larger at once*
- *If both wheels spin forward then the distance will get smaller*

A causal theory can posit hidden (latent) facts needed to make sense of actual observations

For example, an unobserved obstacle could be imagined by the causal theory to explain an inability to move

This is a form of **abduction**, that is, the inference of likely causes from observed effects given general rules

The rules making up a causal theory are expressed as a **logic program** (a collection of deductive rules)

The logic program embodying a causal theory is *discovered* from a small amount of observations, without human intervention

Modeling causality is achieved entirely from active experiencing, not from prior instructions or examples

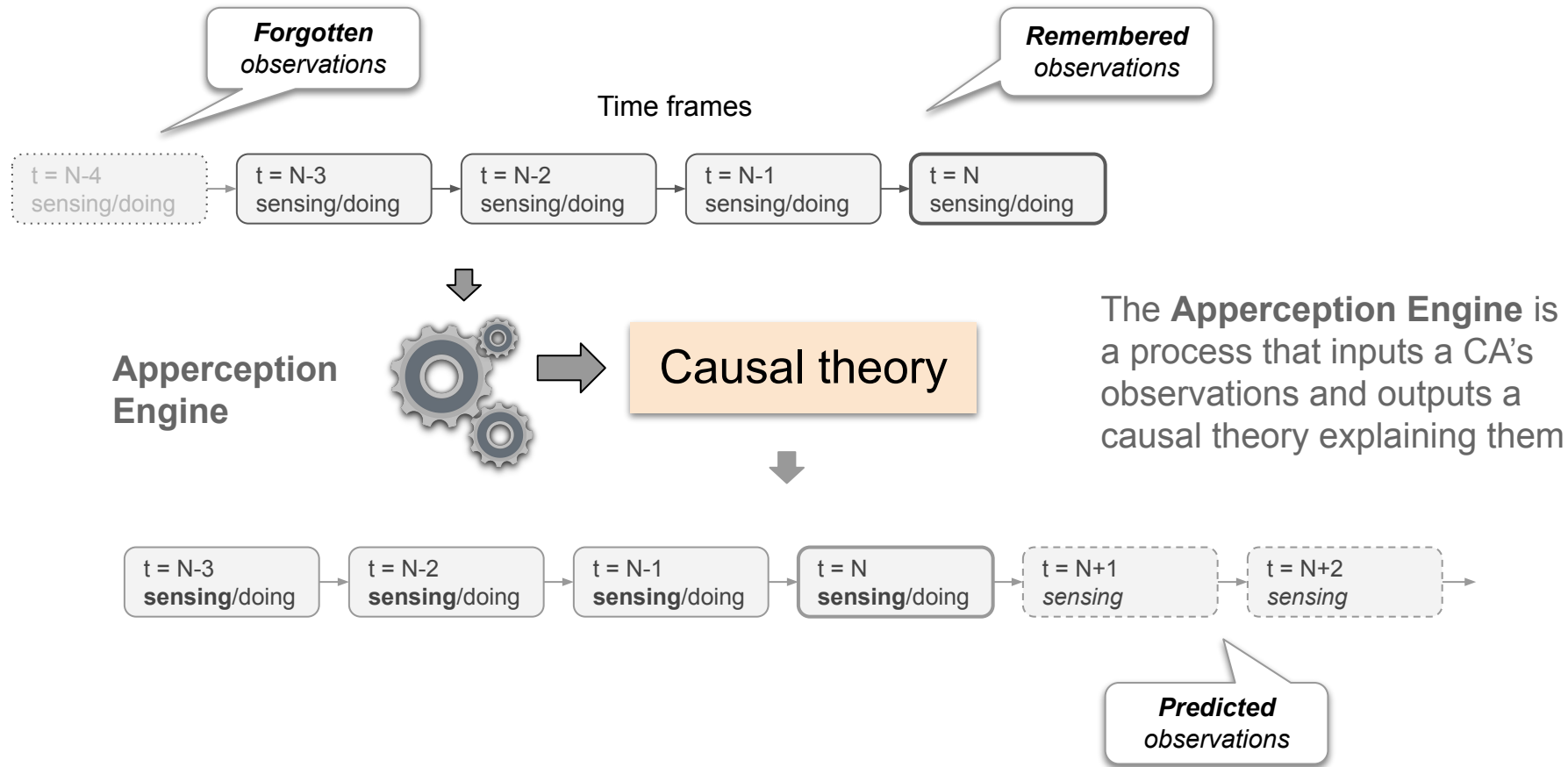
This happens repeatedly over a CA's lifetime

A dynamic CA runs its logic program to predict the next observations, and to do what-if reasoning before acting

The logic program serves as a *generative model*

The Apperception Engine

A dynamic CA requests a logic program (encoding a causal theory) from the **Apperception Engine** by providing it with its latest observations



Dynamic CAs call upon the Apperception Engine when they experience too many surprises using their current causal theories

Or when they fail consistently to predict the consequences of acting upon their umwelts

Finding, from remembered observations, the *simplest* logic program that makes *accurate* predictions about future observations, is a hard *search problem*

The space of all candidate logic programs is extremely large

We need to relax the conditions for success and look for a *good enough* program (simple enough and accurate enough)

We also need to bias the search so it productively ignores most candidate programs

Kant's **Synthetic Unity of Apperception** specifies what makes a causal theory coherent (well-formed) and what gives it explanatory power

It provides strong *inductive biases* that exclude much of the search space

An Apperception Engine is a program that applies Kant's Synthetic Unity of Apperception to find logic programs that encode useful and meaningful causal theories

The original Apperception Engine was conceived and built by *Richard Evans et al.* of Google DeepMind

I re-implemented and optimized it so that good enough causal theories have a good chance to be found in “useful time”

See "Finding causal theories quickly enough - building a responsive Apperception Engine" at <https://zenodo.org/records/15512255>

The Karma System

The project's public name is **Karma**

The design notes and code are in the **karma_system** repo and sub-repos

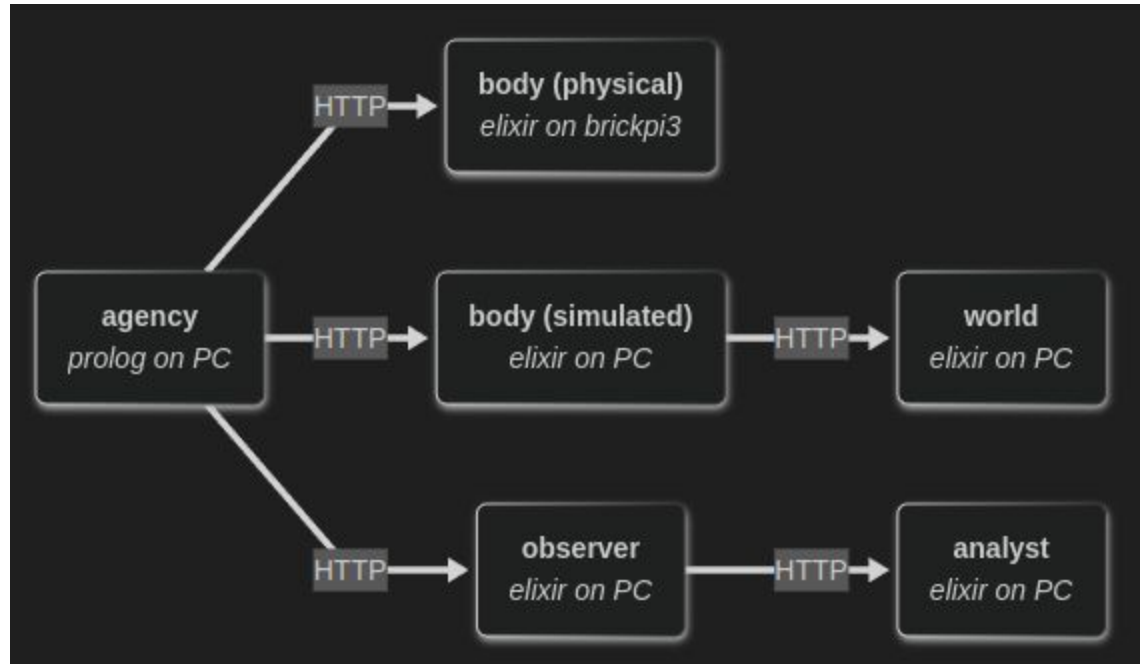
https://github.com/jfcloutier/karma_system

Everything is open-sourced

The Karma sub-repos

- **karma_actors** - Actor Model framework in Prolog
- **karma_apperception** - A practical Apperception Engine
- **karma_agency** - Prolog code implementing agency using actors
- **karma_prolog_utils** - Prolog utility modules
- **karma_body** - Elixir code giving access to the robot's sensors and motors, actual or simulated
- **karma_world** - Elixir code implementing a virtual environment
- **karma_observer** - TBD - Code providing an observer view on the robot's cognition
- **karma_analyst** - TBD - Code facilitating the FEP analysis of data gathered from running the robot

The Karma subsystems communicate over a local network



Progress report

Done - subject to future iterations

- Conceptual and implementation design (karma_system)
- Interface to the robot's sensors and effectors (karma_body)
- Virtual grid world (karma_world)
- Apperception Engine (karma_apperception)
- Runtime platform for CAs (karma_actors)
- Sensor and effector CAs (karma_agency)

In progress

- Dynamic CAs (karma_agency)

To do

- Physical recreation of the grid world
- Observer subsystem
- Analyst subsystem
- Experimental runs and data collection
- FEP data analysis

Parting thought

Should I succeed at programming a robot with felt experiences, who makes sense of its world, and strives to survive

Could there be something it is like to be this robot?



jf.cloutier@activeinference.institute

#sym_cog_robotics on the Active Inference Institute's Discord

https://github.com/jfcloutier/karma_system